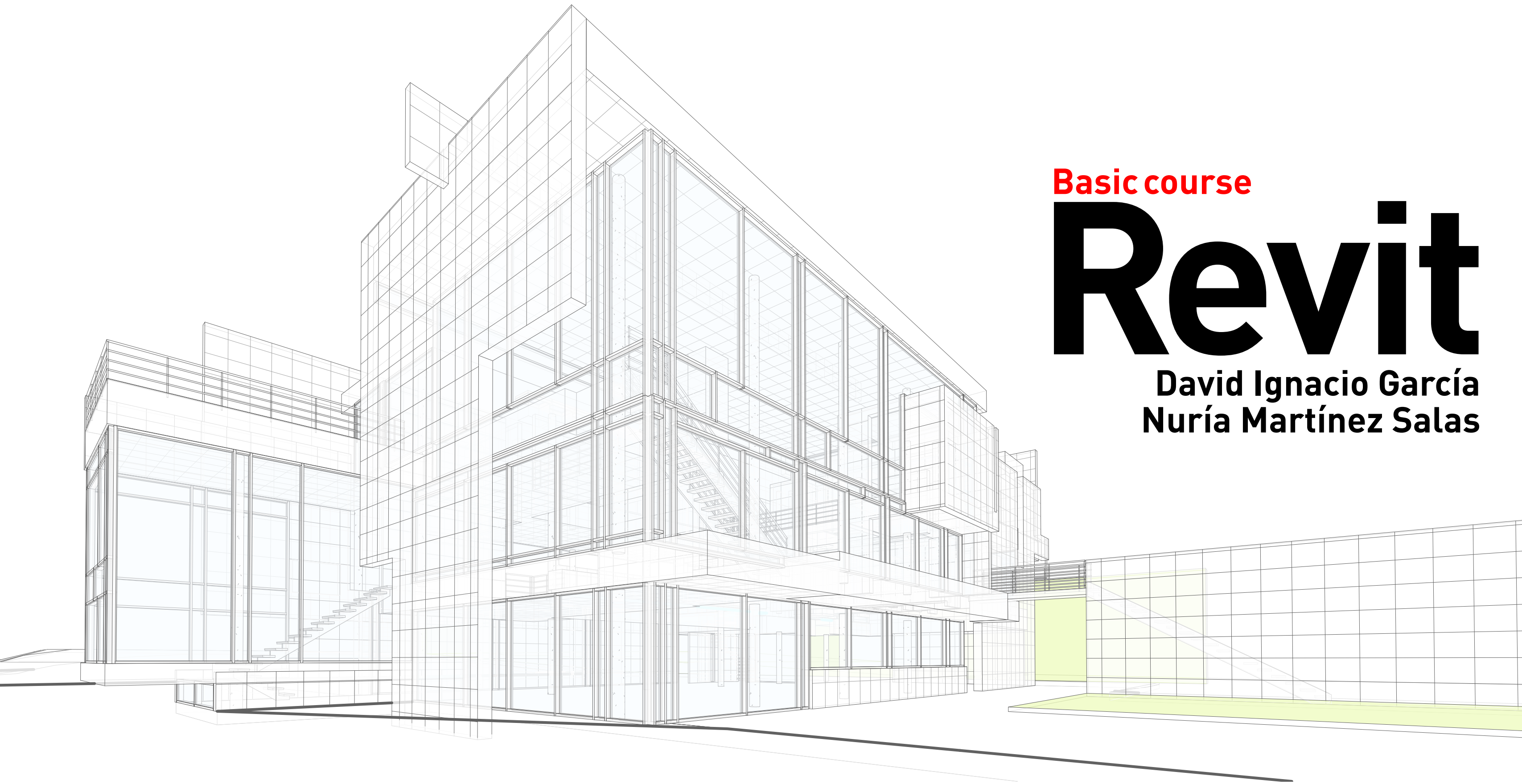




Basic course

Revit

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WHAT IS TOOLS TRAINING?

Tools Training are courses about specific tools that require constant updating. They are specifically designed for the School of Architecture students (degree, postgraduate and vocational training) aiming for a better academic progress in the subjects of their regulated course. The priority areas of focus for the proposed program are software for digital fabrication, parametric design, BIM and use of specific machinery at architecture workshops. The professors will be linked to school and preferably alumni. The learning system will be based in practical classes on workshops. Students will bring their equipment and educational licenses shall be used. Digital documents as well as promotional videos of the courses will be made.

WHAT IS BIM?

BIM (Building Information Modeling) is a design methodology that includes and manages all the information of a building in its life cycle. S-BIM (Structural BIM) collects information about the structure in the same exchange model. This new concept of collaborative work in engineering, construction and architectural design creates a digital representation beyond the planning and design phase of the project, extending throughout the building life cycle, supporting processes that include cost management, construction management, project management and facility operation. S-BIM increases efficiency and the overall quality of the building, and needs specific software to collect all the information. This information includes all the components of the building structure: elements design, analysis, detailing, installation, facility, sustainability, legislation codes, etc... and this must be compatible, in real time, with other building disciplines.

Oscar Liébana

Founder of BIM Lab UEM



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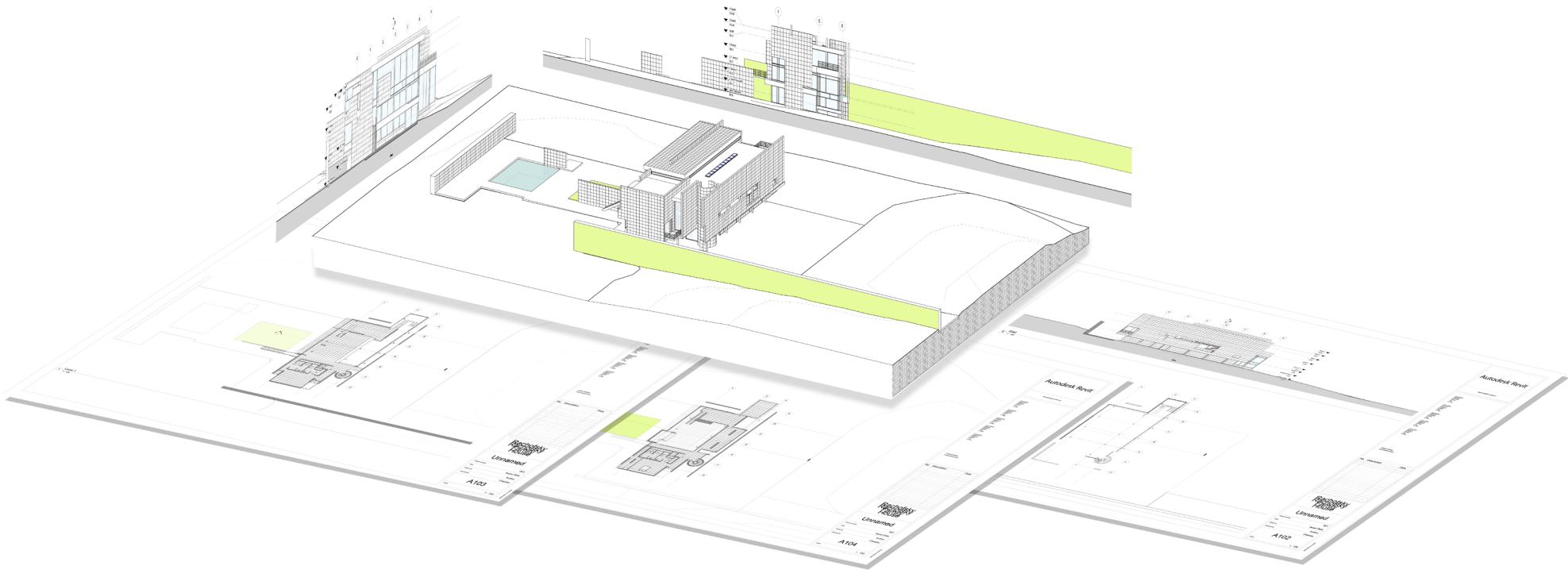
If you are a professional related to the Construction Industry, these three questions below are definitely of great interest to you

What is BIM?

How can it make your professional life easier?

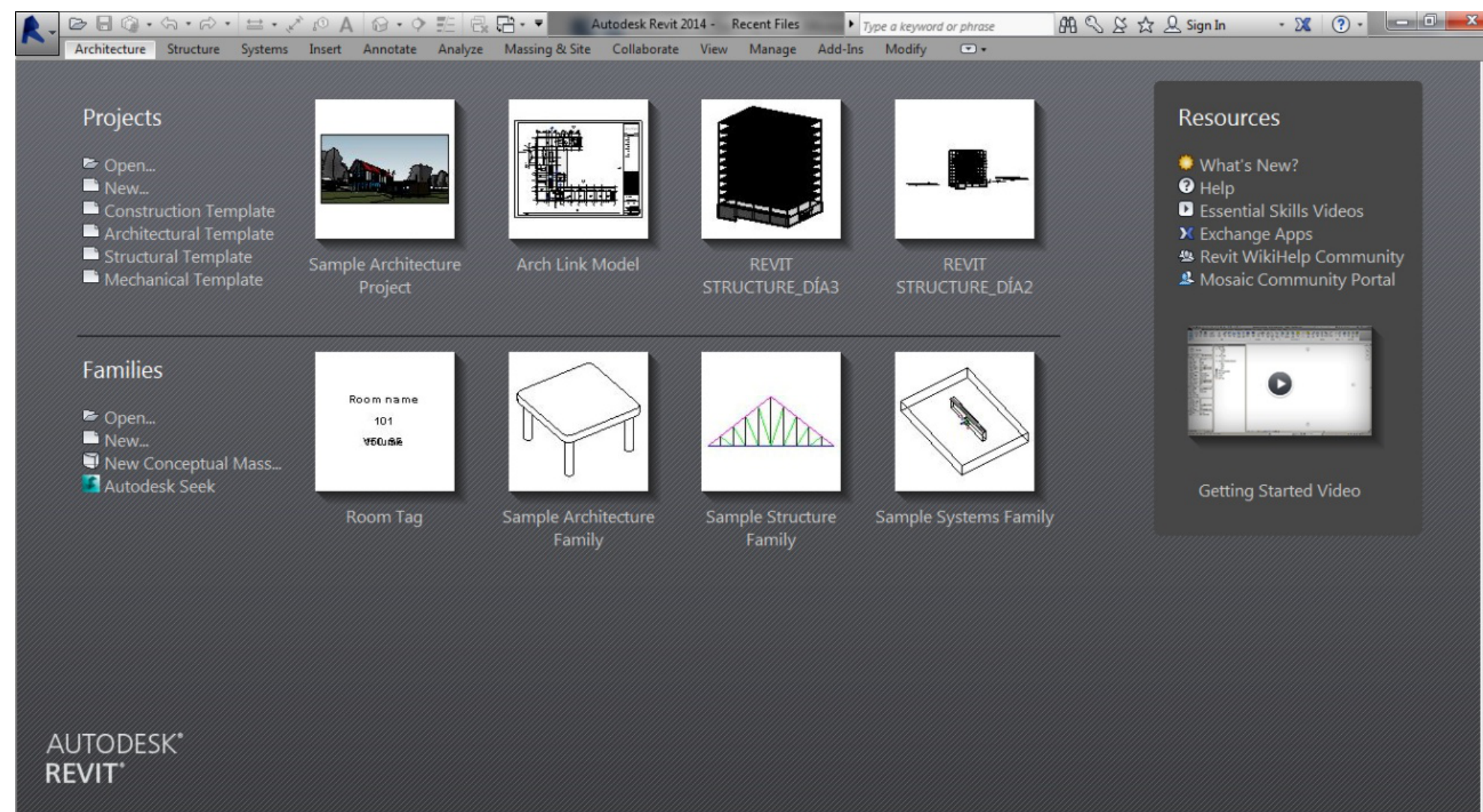
How can it improve your performance and efficiency?

First of all, you should know that BIM in Spain is still relatively new technology in an industry typically slow to adopt change. In other countries however, like USA, UK, Australia, Finland, Sweden or Norway it is widely spread. So, in this time of crisis, this is your chance to become more competitive, efficient and a better professional by mastering something that can make you outstand. Second, you should know that there are different software applications that use the BIM process to function. This document is going to refer to the Autodesk Revit software application, but you need to understand that Revit and BIM are not the same thing. Revit is one of the tools available in the market to help achieve Bim, which is a process that will be described and explained a few lines below.



BIM is a process. Revit is a tool.

In Spain, there are already some Architecture practices and Engineering firms that are nowadays using the software successfully. But even if you were the only one using the software, you would have no problem relating to other design and construction agents, nor with the rest of the industry. The interoperability between Revit and AutoCAD (the most widely spread design application until now) is tremendously easy and simple. Remember that both software applications are designed by Autodesk. The workflow between both is very well thought-out. You could model a building in Revit, and easily export it to AutoCAD without losing a bit of work, and vice versa. Last but not least, keep in mind that this is a whole new world for you. It can improve every aspect of your professional life. But we all know that Rome wasn't built in a day. Even though it will be quite easy for you to learn the software, it takes time to be able to master it and get the most out of it. Be patient. We can assure you the effort is worth it.

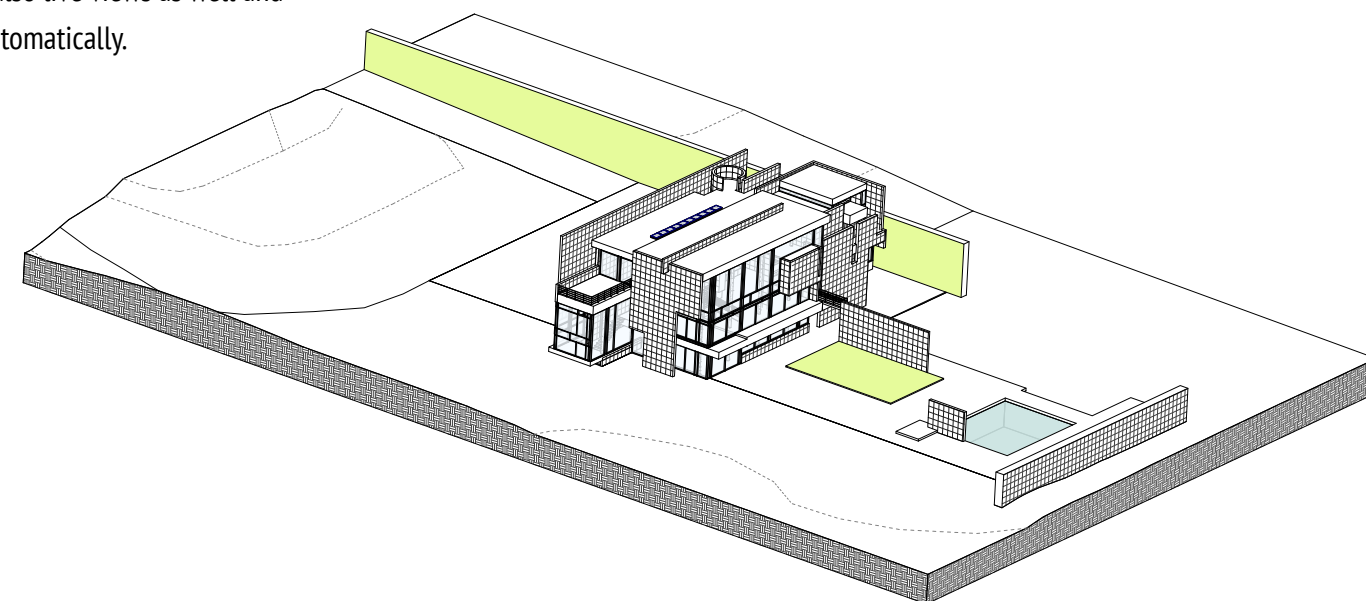


BIM stands for Building Information Modeling, and it is a term that was created a few years back by Autodesk. It was not Autodesk that invented the process for the first time (there were other software applications using this process before) but it was Autodesk who described this process with this term: **BIM**. This is the term that has been widely spread and that is used worldwide to describe this new way of working. BIM is a process of creating **virtual 3D models** that represent building facilities. But these models not only represent the 3D geometry of the buildings, they also contain as much information as needed. What does this mean exactly? Let's explain further. You probably already know that there are many software applications that allow us to represent buildings in 3D. Revit is not the only one that does that. What's important to understand is that Revit is not only about representing geometry in 3D, Revit does much more than that. With Revit, the virtual 3D model is actually a **placeholder** for as much data as we want that model to contain.

In other words, in Revit, the model is also a database, a data store of information. So we can use this data store of information as we need (Revit allows us to generate **live schedules** of any part of our building at any time) and it can also be used internally by the system to do meaningful computations like energy analysis, etc... With the BIM process in general, and in this case, with the Revit software application in particular, we create virtual 3D representations of buildings that reproduce not only the 3D geometry (the form, shape, position, etc...of every element in the building), but also contain as much information about those elements as we want. Carried to the extreme, we could create an **exact virtual representation** of a building containing every single element of that building, with every single bit of data (all the characteristics, properties and parameters) about each element.

Another very important benefit of using the software is coordination. In Revit everything in your project is fully coordinated, so no manual updates are required. Everything is always in sync. This means that you will have **no errors** related to lack of consistency between documents and you will not have to do the same work several times. If something changes, you only have to make that change once. Everything will automatically reflect that change. What does this really mean? It means that in Revit, work that you do in one view is immediately reflected in the others. For example, work that you do in plan is immediately reflected in elevations and sections, and vice versa. But not only that, this behavior is not limited to graphical views, it also **affects schedules** (lists of information about different elements in the project), which are also live views as well and therefore update automatically.

In traditional architectural design and documentation procedures, each drawing conveys only a small part of the whole and can easily get out of synch... In contrast, with the BIM/Revit workflow, all modifications are performed directly in the model. The conventional architectural drawing types like plans, sections and elevations are what in Revit we call **Views**, and these Views are live representations of the model. Since each View is a "window" looking at the whole, the various Views cannot get out of sync, and therefore always accurately convey the current state of the design.

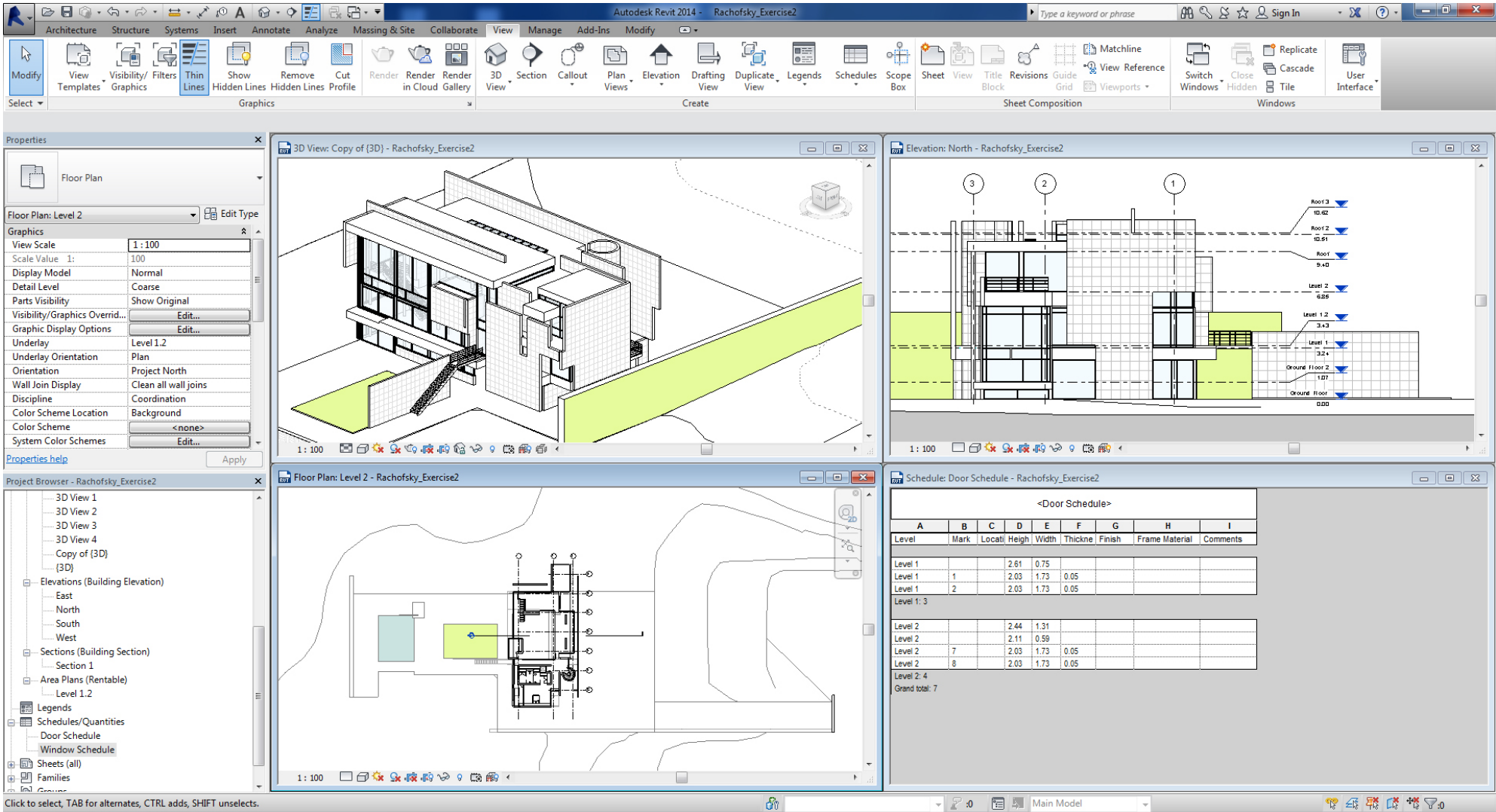


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Building Information Modeling Coordination

In terms of management of a project, Revit gives you a very organized structure that will help you have great **control** over your projects and manage your projects very efficiently.



How can it make your professional life easier?

This question and the next one have partly been answered already. Apart from the fact that you won't have to worry about making mistakes by manually updating your projects or generating errors due to manual computations, Revit will allow you to do basically everything you need when designing a building without having to use any other software application. You can easily get 3D images of your building with all kinds of **graphic display** options or generate renders, solar studies or walkthroughs. This way you won't have to be proficient in different software applications to fully represent your design and make it appealing and easy to understand by clients, you just have to be proficient using one software application.

Another way in which Revit can make your professional life easier is that you will have more time to spend with valuable work because Revit will take care of less valued work. Revit offers you many resources, tools and **ways to elude** having to do **repetitive** and "background" **work**. Also remember that when you are designing in Revit, you are actually modeling in 3D so you won't have to wait until the whole building is fully designed to see a 3D representation of it. You can create your 3D views easily at **any moment** from the very beginning of the project. This way you can see in 3D how your design evolves.

How can it improve your performance and efficiency?

The whole point of Revit is precisely to greatly increase performance and efficiency and to help you spend more time doing creative work and less time doing work that doesn't require as much talent. When working in AutoCAD, we use design software that uses **2D lines** to represent buildings. This is the same concept as drawing with a pencil on a sheet of paper, except that we are doing it with a computer. It is true that AutoCAD has also some smart features that couldn't be achieved when drawing by hand, but in both cases we are representing our building through 2D lines. Revit is software specifically meant for **designing buildings** (also their Structure and Systems) so everything is conceived thinking of that purpose. In Revit we don't use 2D lines to represent elements, we use an accurate virtual 3D model representation of those elements.

With AutoCAD, if we want to represent a Wall in plan, for example, we have to use **two parallel** 2D lines, which is actually a very abstract way of representing a Wall. On the contrary, in Revit, if we want to represent a Wall in Plan, we use an element that is actually virtually reproducing that wall in 3D, and that has not only the appropriate size, form and shape (geometry), but also all the characteristics and properties (information) of that wall. To see it in plan, we just have to cut that object with a **horizontal plane**. So the fact that we are not working with abstract 2D lines to create different representations of the building, that are actually **disconnected** to one another (like we did in traditional architectural design), but rather virtually modeling our building, is actually a step forward towards efficiency.



Revit element Hierarchy

At this point, you have already discovered that the way we work using software like Revit is quite different from the way we work using software like AutoCAD.

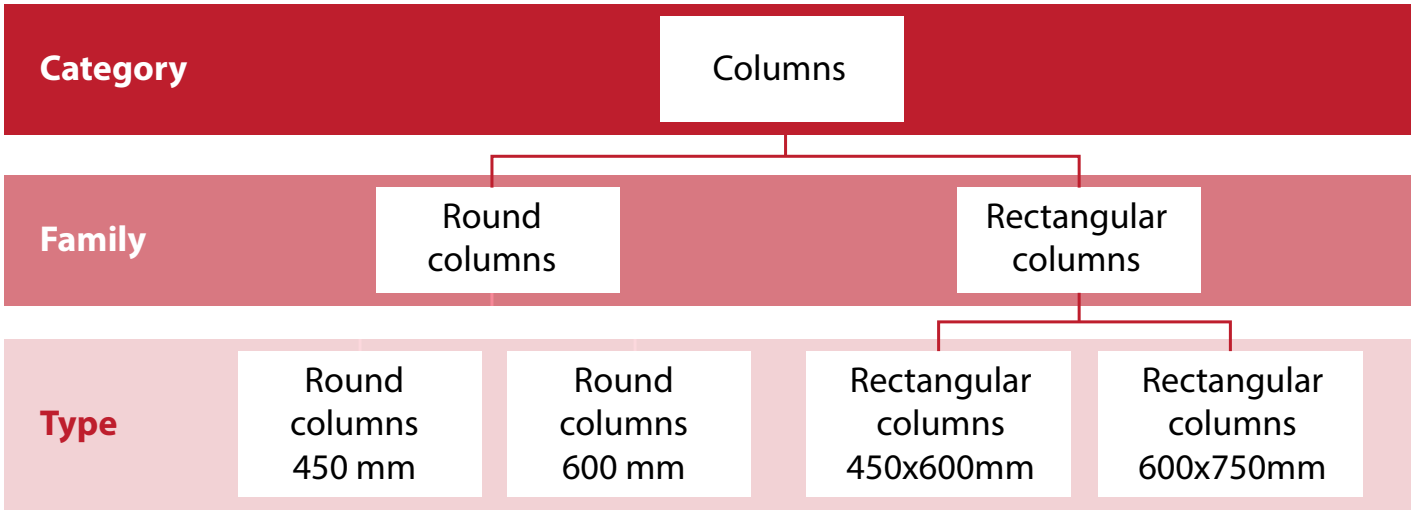
The end goal in Revit is to have a virtual 3D representation of a building that can be as perfect and accurate as we want. This accuracy depends on the **Level of Development** (LOD) that we want to give to that model. Let’s use a simple example. If we want to represent a wall with a low Level of Development, we could just use a **Generic Wall Type** that basically only represented the size, shape and form of that wall (geometry). But if we wanted to represent a wall with a high Level of Development, we could create a Type of wall that reproduced not only its geometry but also contained all the **characteristics** of that wall (information). This information could include the **construction layers** defined with their materials, functions and specific properties. This is how we could have and exact virtual replica of the real wall.

Category

Revit uses Categories to “classify” elements. They are like “big buckets” that Revit uses to organize elements. Categories are **built into** the software. We cannot create new categories nor delete any. Some examples of Revit Categories are: Walls, Roofs, Floors, Stairs, Doors, Windows, Furniture... Each category has its **own properties** and behaviors. The fact that an element belongs to a certain category, means that it will have a certain set of properties and will behave in a **certain way**.

For example, the fact that an element belongs to the Wall Category, is letting the software know that that element can be a host for doors and windows, among other things. On the contrary, if an element belonged to the Floor Category, for example, it could never be a host for a door.

Example:Wall: Basic Wall: Brick Wall



Category : Family : Type

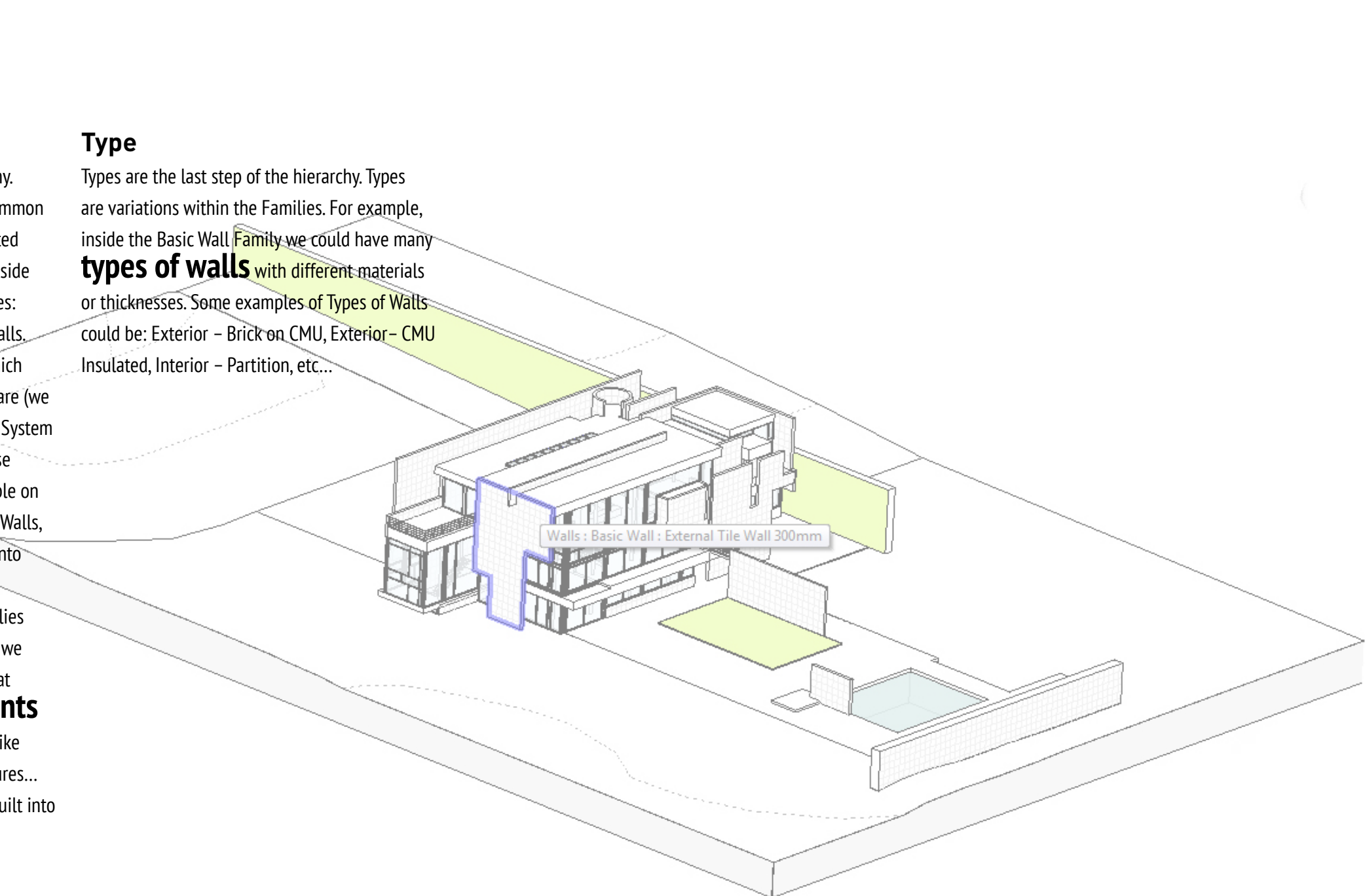
This is a three step **object hierarchy** that all elements, both model, and annotation, share in common. Every element in Revit belongs to a type, that type belongs to a **Family** and that Family belongs to a **Category**. For example, we could have a Brick Type of wall that belongs to the Basic Wall Family that belongs to the Wall Category.

Revit element Hierarchy

Families are the next step of the hierarchy. A family is a group of elements with a common set of properties (parameters) and a related graphical representation. For example, inside the Wall Category, there are three Families: Basic Walls, Curtain Walls and Stacked Walls. They are **System Families**, which are Families that are built into the software (we cannot create new ones nor delete any). System Families are the ones that represent those elements that we typically would assemble on **construction site**, like Floors, Walls, Roofs, Stairs... System Families are built into the system, But there are also Loadable Families, which, on the contrary, are families that we can create ourselves exactly like we want. Loadable Families are the ones that represent **building components** that we typically buy to a manufacturer, like Doors, Windows, Furniture, Lighting Fixtures... **Loadable Families** are not built into the system.

Type

Types are the last step of the hierarchy. Types are variations within the Families. For example, inside the Basic Wall Family we could have many **types of walls** with different materials or thicknesses. Some examples of Types of Walls could be: Exterior – Brick on CMU, Exterior – CMU Insulated, Interior – Partition, etc...



Revit element groups

All of the elements in the software could be grouped into:

Model elements

Model elements are the ones that represent something that is real in the building, like walls, doors, stairs, windows...They are 3D objects. A model element is a live object that we can visualize from all views.

Annotation elements

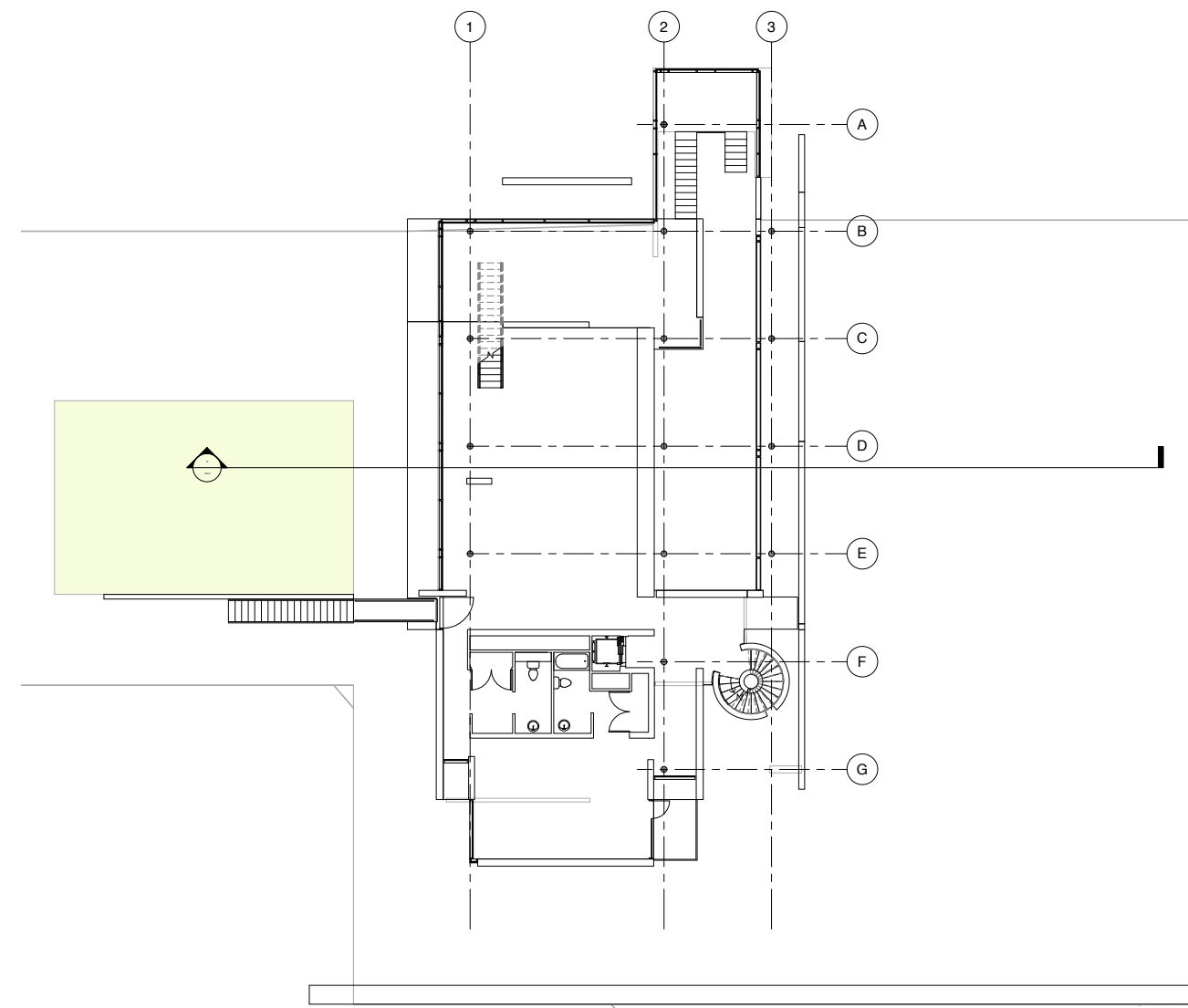
Annotation elements are things that are not real, they only describe objects in a set of drawings. They are 2D objects. The annotation elements are what we call view specific. If we place them in one view, they will only see them in that particular view.

View Elements

View elements are the elements that can take us to our project Views like Elevations and Sections.

Datum elements

Datum elements are organizational elements like Levels and Grids. They serve as references to be able to place our model elements.



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INTERFAC

Understanding the different “flavours” of Revit

At this point you should know that there is a variety of “flavors” of Revit:

Depending on the “flavor” you have, you will find that some commands in the interface are different, but the organization is the same. It is **very important** that you learn the names of each part of the interface. Not only because we will refering to them later in this document, but also because when you look for an answer on the internet about some Revit question, you will need to know how things are called to understand the answer they are trying to give you. Let’s take a look:

Revit, which is the full product that includes functionality for all disciplines: Architecture, MEP, and Structure.

Revit MEP, which includes the Mechanical, Electrical and Plumbing functionality.

Revit Architecture, which includes only the architectural functionality.

Revit Structure, which includes the Structural Functionality for Structural Engineering.

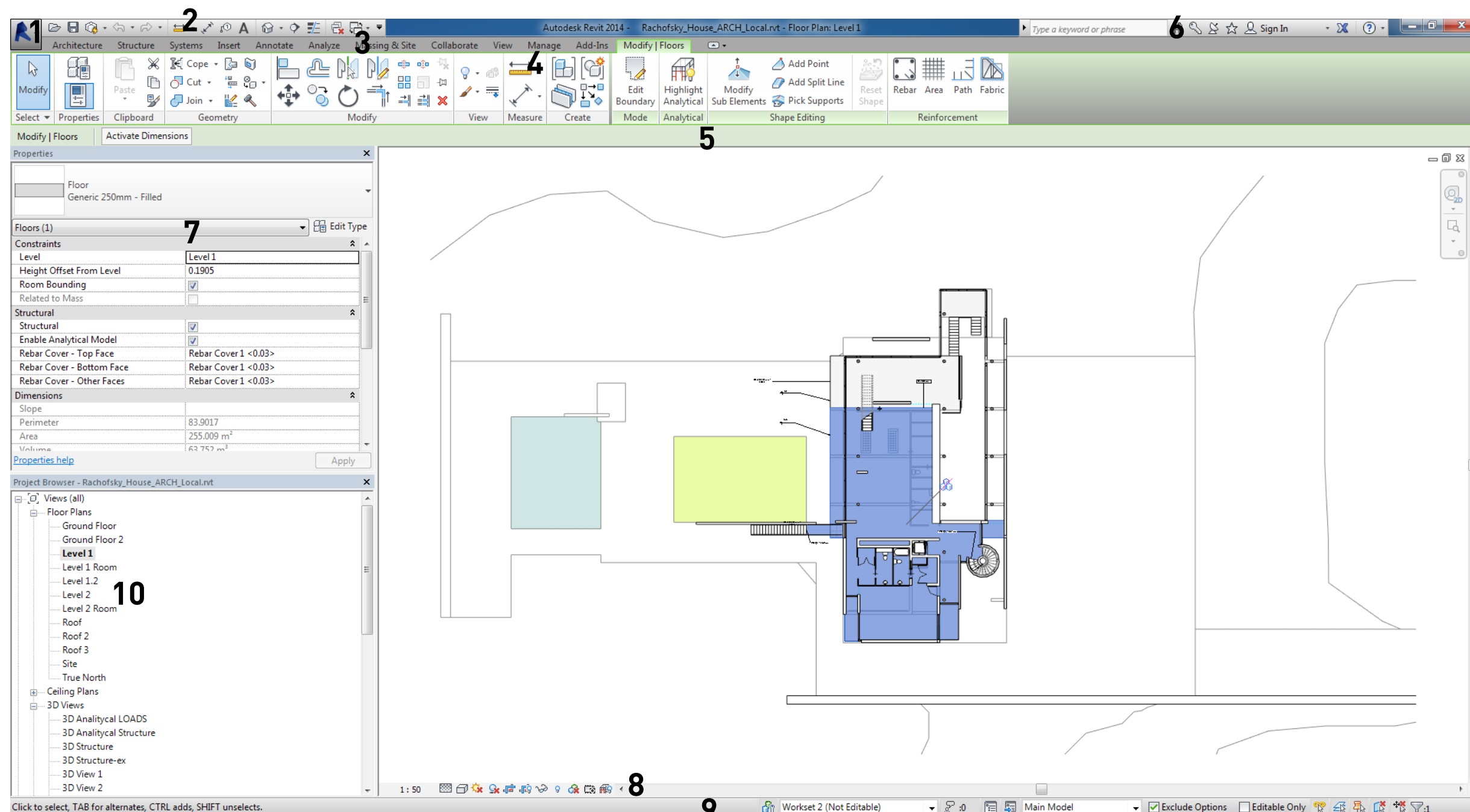
Revit LT, which is a limited version of Revit Architecture.



1 The **application menu** provides access to common file actions, like New, Open, Save, Print, Export...

2 The **Quick Access Toolbar** gives you quick access to certain commands that are frequently used.

3 The **Ribbon** is the primary interface for most commands in Revit. In the Ribbon you have various tabs, each tab is broken into panels and on each panel there is a collection of buttons. It provides all the tools necessary to create a Project.



4 The **Context ribbon tab** appears on the ribbon as you create or modify elements. It will give you different context buttons depending on what you want to create or modify.

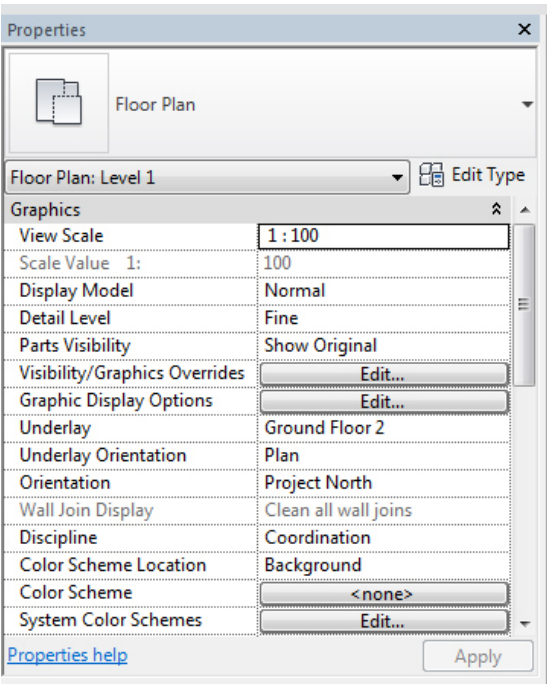
5 **Options bar** is also context sensitive. If you run a command the options bar will populate with several options depending on the command you are running.

6 The **Info Center** gives you access to several internet resources.



7 Properties Palette

The Properties palette is a dialog where you can view and modify the properties of elements in Revit. Most elements in Revit have both Instance Properties and Type Properties. From the Properties Palette you have access to both. Let’s explain the difference between these two. We have mentioned already that all elements in Revit belong to a Type (every Type to a Family and every Family to a Category). Well, every Type of element has Type Properties and every single element that belongs to that Type will have those same Type Properties. On the contrary, Instance Properties only belong to a single Instance of a Type. If you change a Type Property, you are changing that Property for all elements that belong to that Type. If you change an Instance Property, you are only changing that property for that particular Instance. The Properties palette is context sensitive. Depending on what you want to create or what element you have selected (you have to select first to be able to modify), you will see different sets of Properties. The Properties Palette is organized this way: At the very top, there is the Type Selector and beneath it, there is the Instance Properties. We can also access the Type Properties by using the Edit Type button.



8 View Control Bar

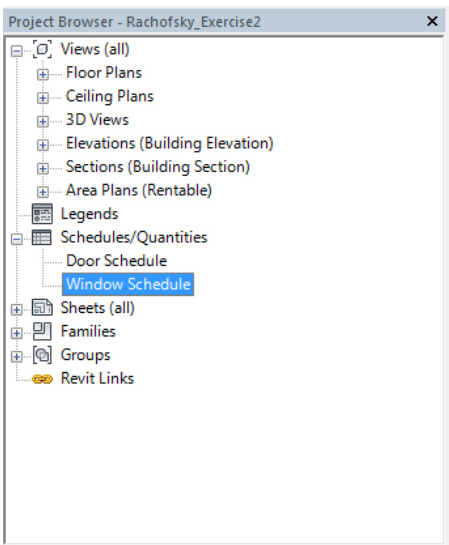
It provides quick access to functions that affect the drawing area: Scale, Detail Level, Model Graphics Style, Shadows On/Off...

9 Status Bar

It provides tips or hints on what to do when you are using a tool. When you are highlighting an element, it displays the name of the category, family and type of that element. It also gives access to Worksets and Design Options.

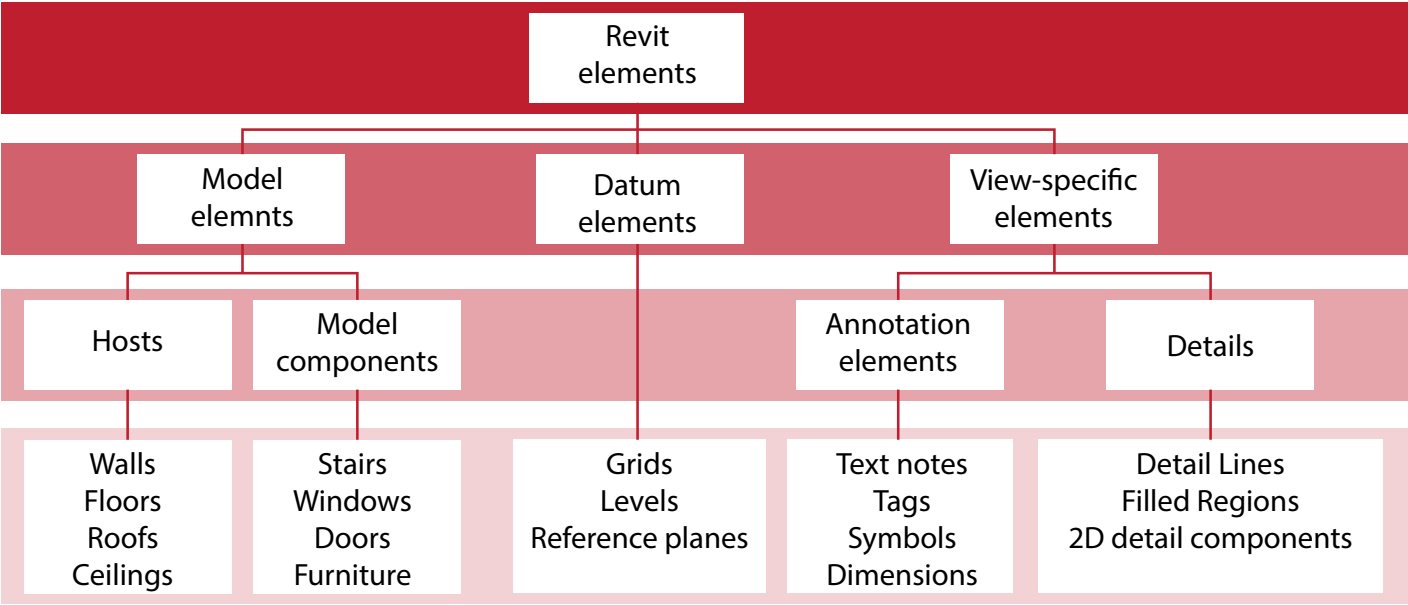
10 Project Browser

The Project Browser functions like a table of contents. A Revit Project is a complete virtual building model. Each representation of a project (Plans, Sections, Elevations...) is referred to in Revit as a View. Views are listed in the Project Browser and we can easily access them from there. The default grouping is divided into typical drawing types: Floor Plans, Ceiling Plans, 3D Views, Elevations, Sections...Beneath that, there are other major groupings like Legends, Schedules and Sheets. The project Browser will help you have a good control over your project and efficiently manage your project deliverables.



Revit Architecture

This document will be basically focused on Revit Architecture, which includes only the architectural functionality. We will be explaining concepts based on an Architectural Project, but keep in mind that you can use Revit not only to model Architecture, but also to model the Structure of a building and it's Mechanical, Electrical and Plumbing equipments.

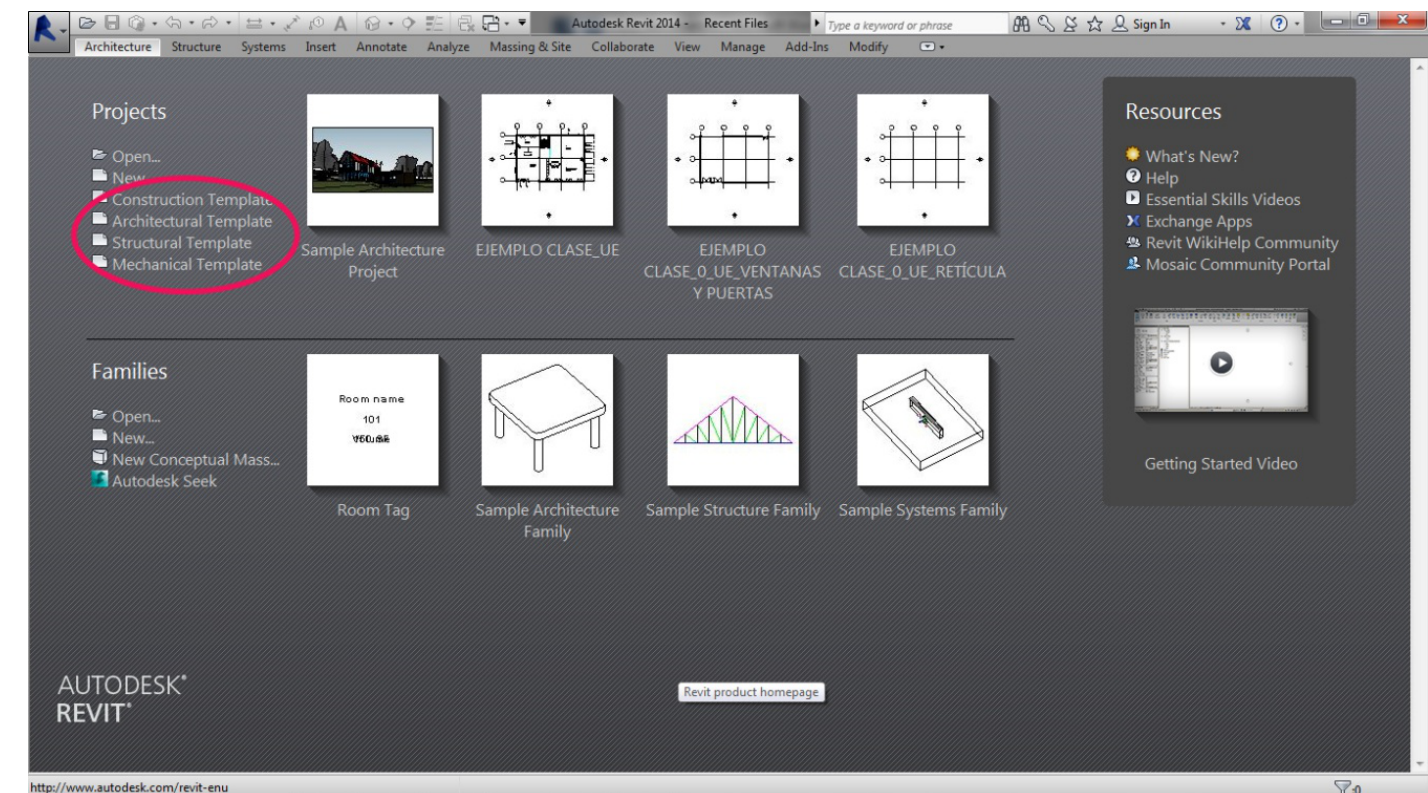


Starting a Project

Creating a new project from a template

All projects in Revit should begin with an appropriate **Template**. The better the Template, the faster you can go when working in Revit. A good Revit Template is essential in improving efficiency, quality, clarity and consistency in your projects. There are a lot of things that can be **pre-built** and put into the Template to get you started. You can have your own custom Template or you can choose from the ones that come with the software. Templates contain a set of repeated elements and are created to standardize settings across the final documents. Using a Template allows for less time with background common and repetitive work.

Templates also provide a level of consistency across all projects, which is interesting from a **marketing perspective**. The Revit Templates continually evolve. Every time you do some background work in one project that could be used in the next one, you should include that work in the Template. Templates also evolve with the office standards. It is important to keep the Templates up to date.



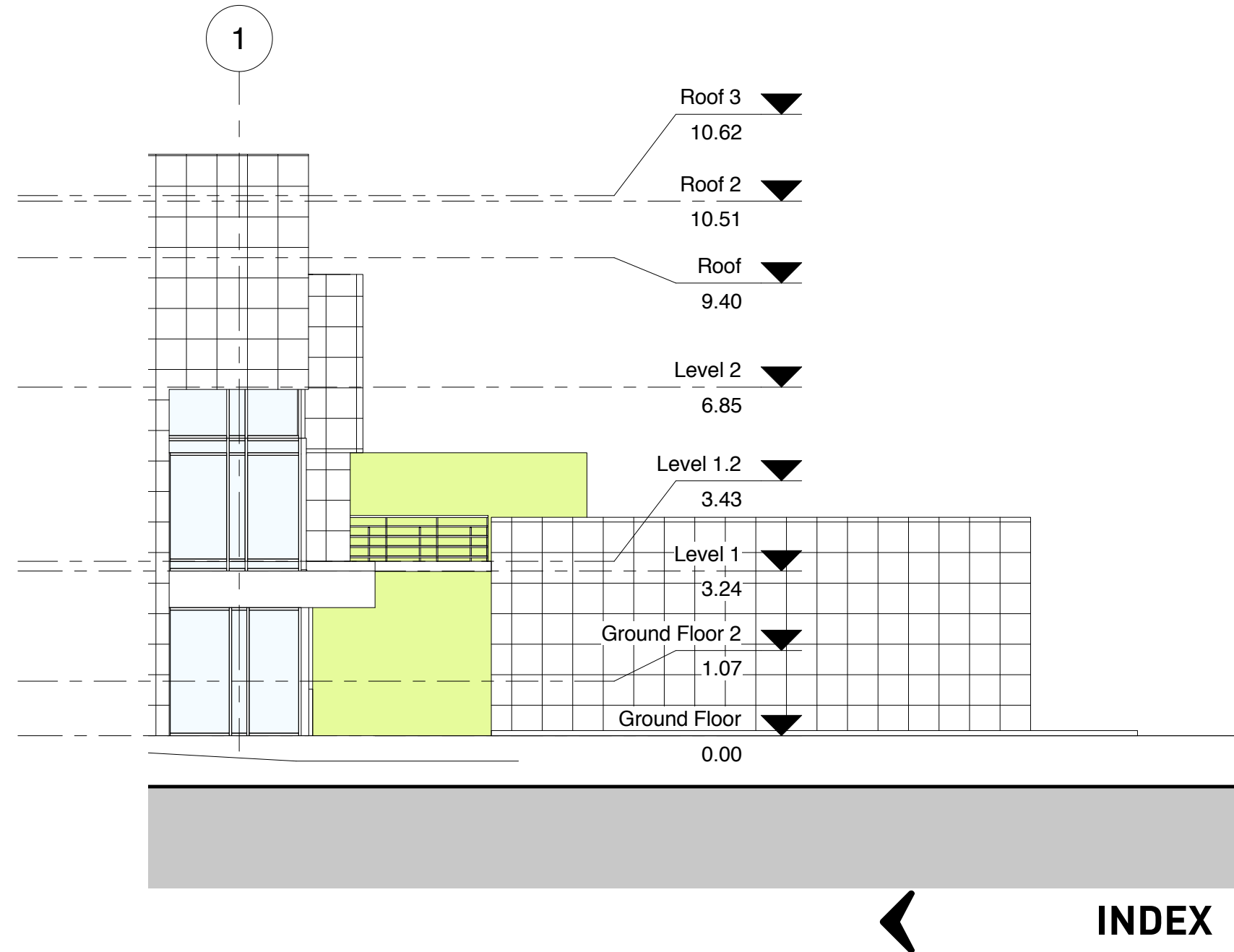
RECENTS FILES screen showing PROJECTS TEMPLATES links



Starting a Project

One of the first things you should do when starting a project is to **create the Levels**. Levels are one of the primary organizational and structural constructs in a Revit project. Levels are **datum elements**, which are organizational elements that establish some known location in your project. We can see levels in elevation views. They act like **horizontal planes** cutting through the 3D model of your building at a particular height.

Adding Levels



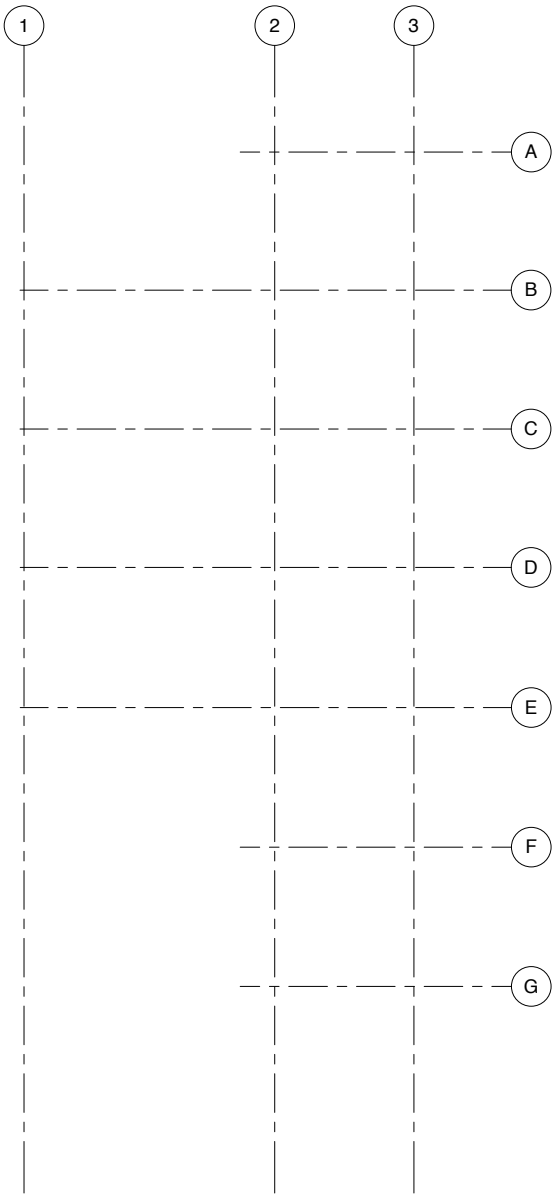
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Starting a Project

Adding Grids

Like levels, grids provide organization and structure to your project. Where levels were horizontal datum elements cutting through the model at a certain height, grids are **vertical datum elements** that cut the model vertically. Grids are used typically to locate where the columns occur in your projects but they also provide **key points** of reference planes for various things.



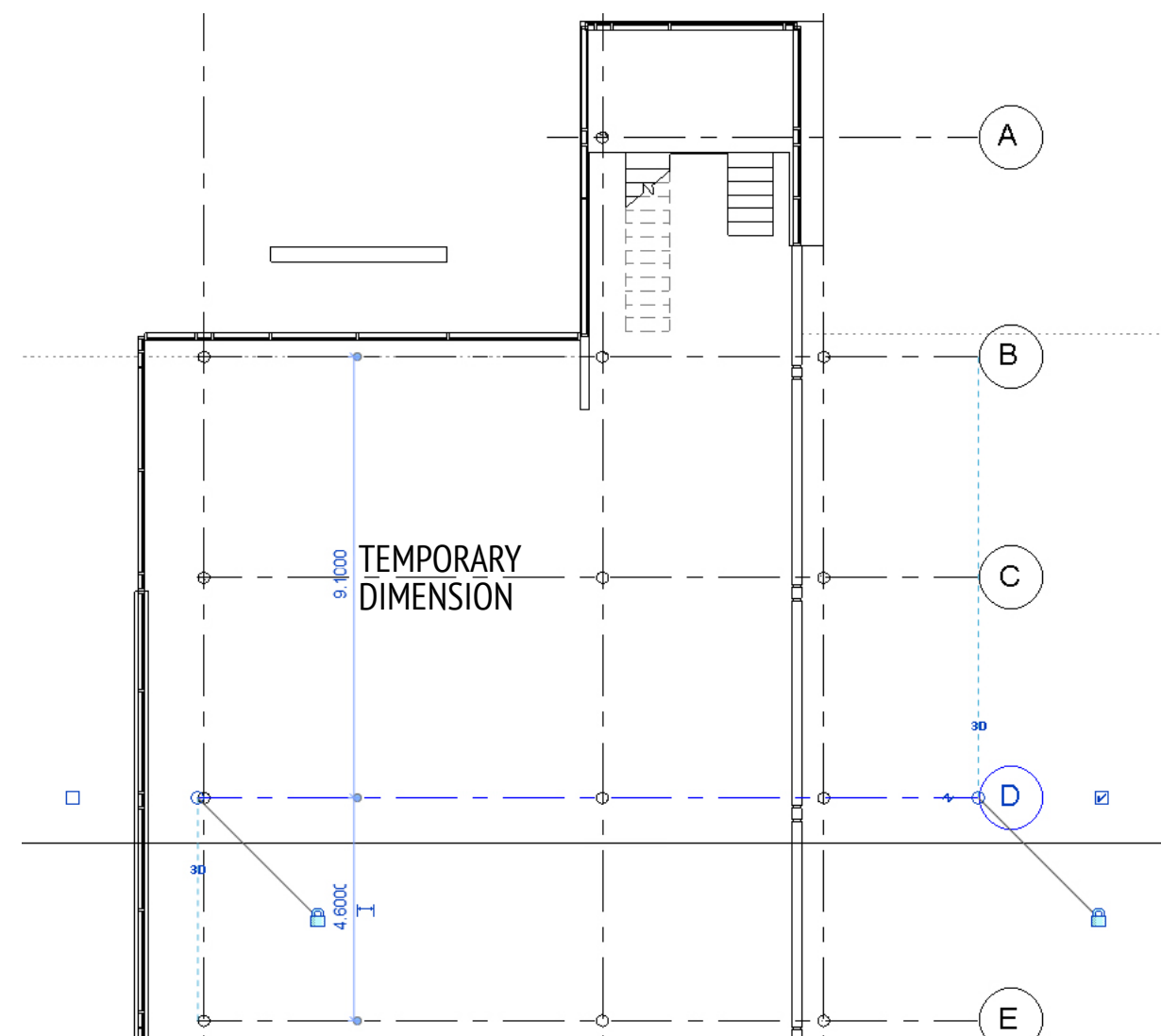
Starting a Project

Refining a layout with temporary dimensions

A very important basic concept to understand in Revit is the way that precision is approached. In Revit, what you typically do is you place objects in an approximate location and then you refine their placement over time. **Temporary dimensions** are something that we use to take our roughly placed objects and modify their positions to a more precise location.

But is there really a need for Temporary Dimensions?

The name Revit comes from “Revise it”. When you are designing, you keep refining and revising your design continuously until you reach the final desired solution. At first, you have a vague idea of what you want and over time it becomes clearer on what the best solution is. Revit understands this **creative process**. You can start by placing objects at an approximate location and then revise and refine their position over time to make the design more and more **precise** as you know more about your project.



Modeling Basics

When we are starting a project, we probably already know **some information** about that project: Client Name, Issue Date, Project Name, Location...You can enter this information at any time, but if you already know it at the beginning of the project you might as well enter it at an early stage. This information will be **stored** in the Project to be used by the software at the appropriate time. For example, the Project Location will be useful when you do Solar Studies and the Project Date, Name, etc...will be used by the software when you use **Title Blocks** to create your Sheets.

Project Properties

Family: System Family: Project Information

Load...

Type:

Edit Type...

Instance Parameters - Control selected or to-be-created instance

Parameter	Value
Identity Data	
Organization Name	Universidad Europea de Madrid
Organization Description	School of Architecture
Building Name	Rachofsky House
Author	Richard Meier
Energy Analysis	
Energy Settings	Edit...
Other	
Project Issue Date	Issue Date
Project Status	Built
Client Name	Rachofsky Family
Project Address	Edit...
Project Name	Rachofsky House
Project Number	001

OK

Cancel

Entering Project Information

Location Weather and Site

Location

Weather

Site

Define Location by:

Internet Mapping Service

Project Address:

8605 Preston Road, Dallas, Texas 75225, EE.UU.

Search

Map

Satellite

Google

Use Daylight Savings time

Aceptar

Cancelar

Ayuda

Modeling Concepts

Way to proceed

The Ribbon provides all the tools necessary to model a project. This is the basic process you follow when you work in Revit.

1 Go to the Ribbon and look for the tool you need. The Ribbon is structured in Tabs, then Panels and then Buttons (tools).

2 Once you have found the tool you need, click it, and immediately you will see that the Context RibbonTab activates. In this Tab you will find some specific contextual buttons that only appear when you click that specific tool.

4 Look at the Properties Palette and choose the Type of object you want to create (or create a new Type) and give the Instance Properties the values you want (some of them related to the position of the object).

3 Check what options you have available in the Options Bar. It will be also offering you some other contextual options.

5 Check the Status Bar, the software will be giving you instructions about what to do when using a tool.

7 When you are done, click the Modify Button to exit that tool (or Esc Key twice) or if you want to model another instance using that same tool, click the Esc key once to activate the Properties palette and be ready to model a new instance.

6 Start drawing in the Drawing Window.



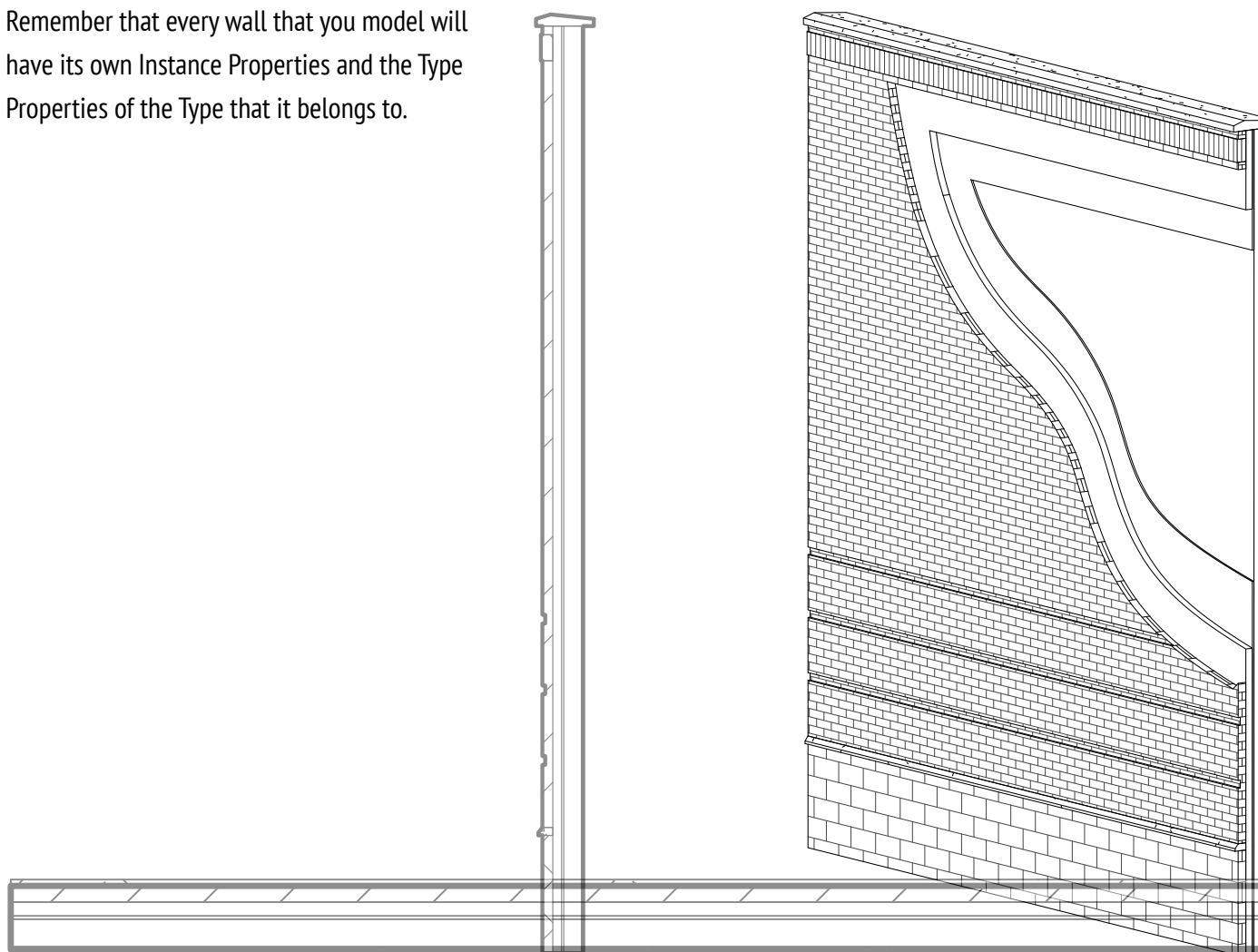
Basic Model Elements

Walls

Walls are perhaps the most basic component of any building project. You can model walls in Plan Views or 3D Views. Remember that one of the Categories that exist in Revit is the **Wall Category**. Inside the Wall Category there are three System Families: Basic Wall, Curtain Wall and Stacked Wall. A **System Family** is a family that is built into the system. System Families cannot be changed. We cannot create new System Families nor delete any. Every wall in your project will belong to one of these three Families. On the contrary, you can create **new Types** that belong to any of those Families. For these new types of walls you can define the construction layers you want, with the thickness, function, materials and properties that you find convenient. Walls are level hosted elements, meaning that when you model a wall, you will have to indicate its Base Constraint and Top Constraint, referred to the Levels.

These two properties (Base and Top Constraints) are **Instance Properties** and therefore have to be defined instance by instance. With the Top and Bottom constraints, we actually attach the top edge of the wall or the bottom edge of the wall to a particular level. This is not a one-time operation, it is a **constraint**. If these levels were to move, the top edge and bottom edge of that wall would follow along with the levels to which they were constrained to. This is a really powerful feature in the software. As your design changes over time, if your levels move, all of the walls that are associated with those levels will move accordingly.

Remember that every wall that you model will have its own Instance Properties and the Type Properties of the Type that it belongs to.

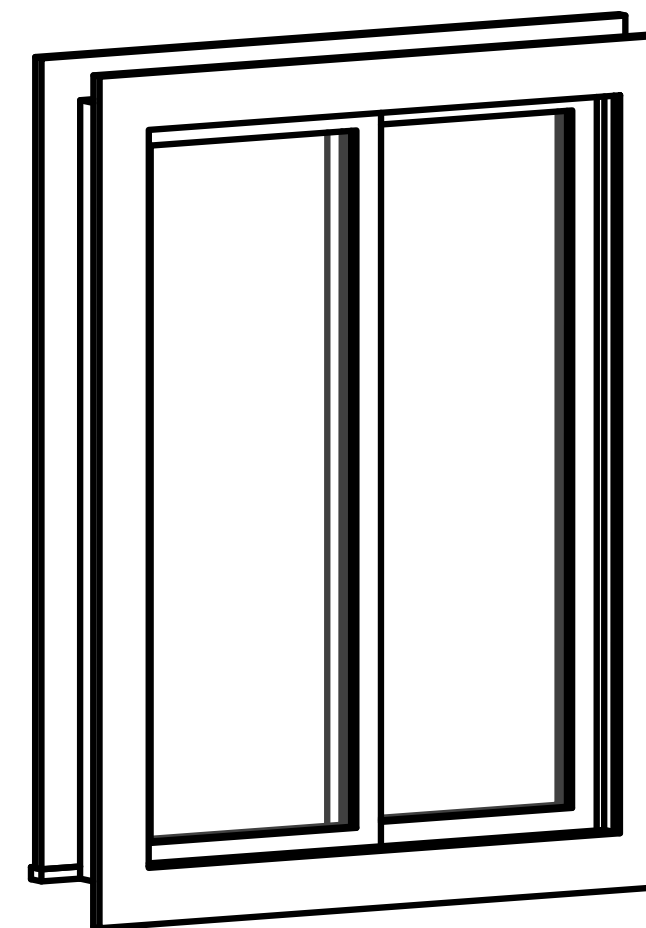
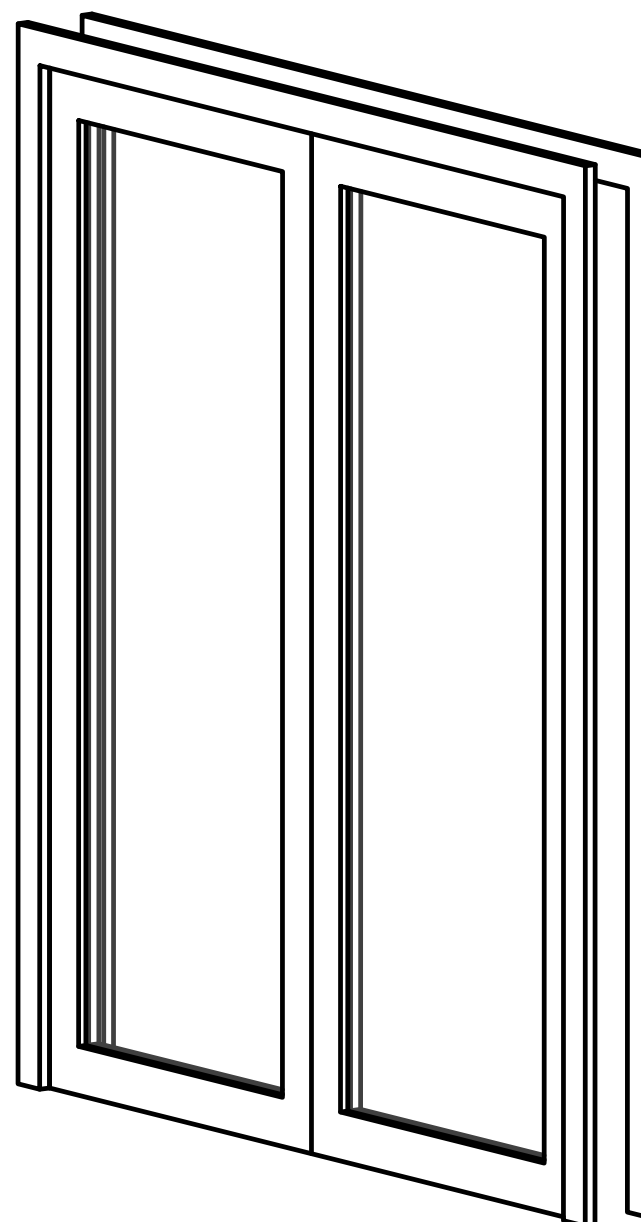


Basic Model Elements

Doors and Windows

Once you have placed walls in the project, you can then place **doors** and **windows**. For the Door Category and Window Category there are no System Families built into the software. In this case you can create your own Families in the **Family editor** and load them into your projects. If you don't want to create our own Families (or you don't need to), Revit offers you some "Out of the Box" Loadable Families that you can use. There is also online content to download all types of Families, like **Autodesk Seek** for example. Some manufacturers are also starting to offer their products as **Revit Families**, ready to be used in your Projects.

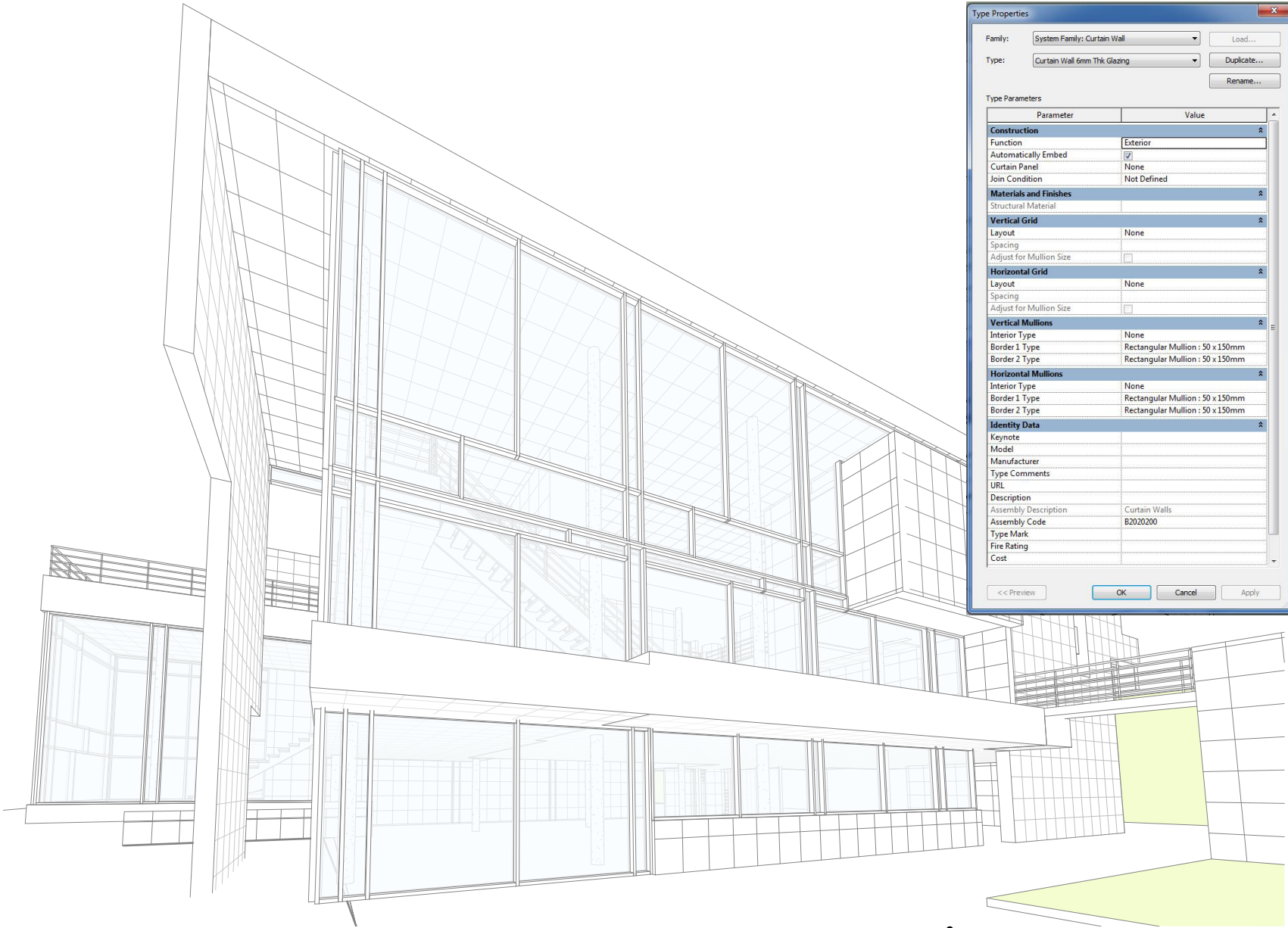
Here, we are talking about Door families and Window families, but this applies to any other kind of Family. When you are going to place a Door or a Window in your project you should notice that it will only let you do it when you place your cursor over a wall. Doors and Windows need walls to exist. They are wall **hosted elements**.



Basic Model Elements

Curtain walls are the third and most complex form of a Wall System Family. Curtain walls have grids, mullions and panels. Remember that Curtain walls are just another wall family so you can draw Curtain Walls the same way you draw Basic Walls. You can have control over the placement of **grids** and the type of mullions and panels. You can define these settings at the **Type level**, which is the most common way to do it, or you can design your own free design for one curtain wall in particular. Curtain Walls can have a **Property** that allows them to automatically embed into other types of walls, but even if this feature was not one of the Curtain Wall Properties, you can always manually embed a Curtain Wall in to Basic Wall or a Stacked Wall.

Complex Walls



Type Properties

Family: System Family: Curtain Wall Load...

Type: Curtain Wall 6mm Thk Glazing Duplicate... Rename...

Type Parameters

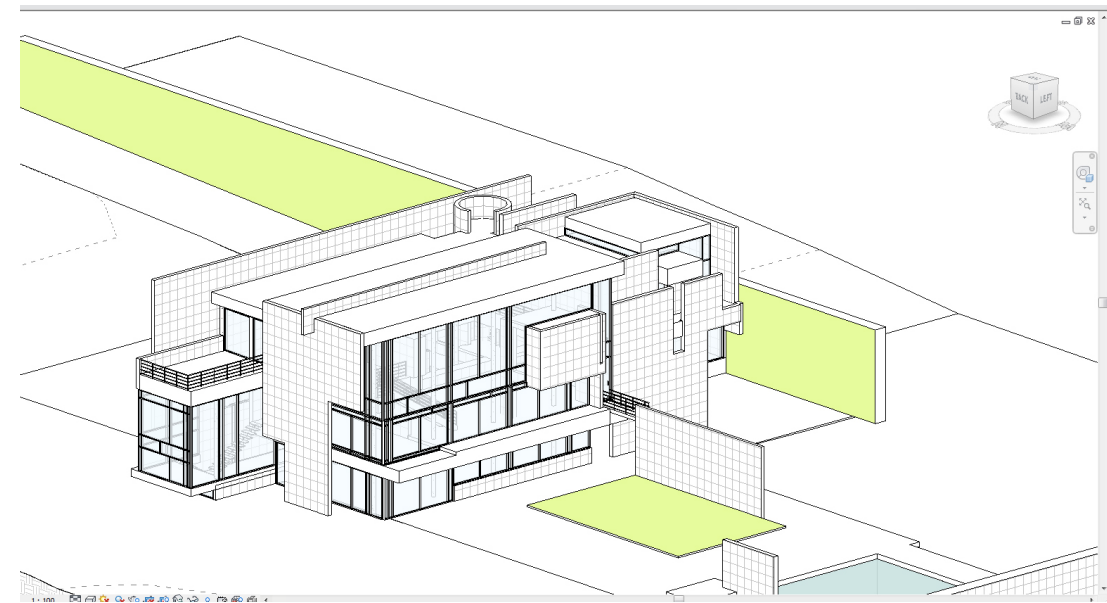
Parameter	Value
Construction	
Function	Exterior
Automatically Embed	☒
Curtain Panel	None
Join Condition	Not Defined
Materials and Finishes	
Structural Material	
Vertical Grid	
Layout	None
Spacing	
Adjust for Mullion Size	☐
Horizontal Grid	
Layout	None
Spacing	
Adjust for Mullion Size	☐
Vertical Mullions	
Interior Type	None
Border 1 Type	Rectangular Mullion : 50 x 150mm
Border 2 Type	Rectangular Mullion : 50 x 150mm
Horizontal Mullions	
Interior Type	None
Border 1 Type	Rectangular Mullion : 50 x 150mm
Border 2 Type	Rectangular Mullion : 50 x 150mm
Identity Data	
Keynote	
Model	
Manufacturer	
Type Comments	
URL	
Description	
Assembly Description	Curtain Walls
Assembly Code	B2020200
Type Mark	
Fire Rating	
Cost	

<< Preview OK Cancel Apply

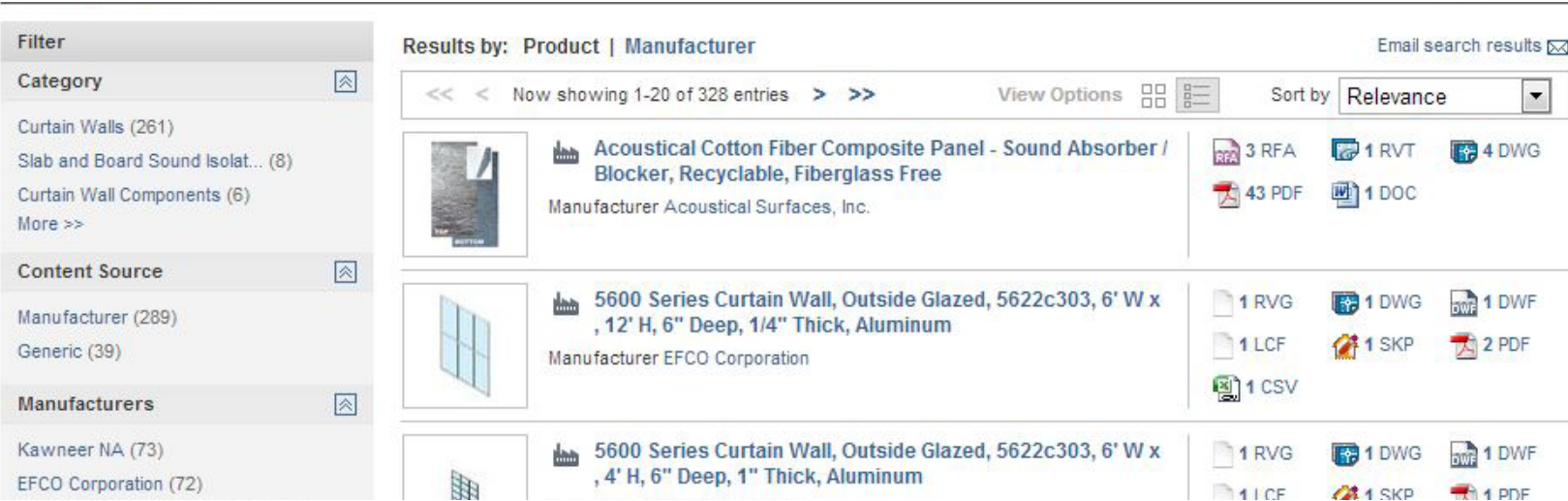


Basic Model Elements

We have already talked about some components. In Revit, Components and Families are the same thing. Doors and Windows, for example, have their dedicated tool in the Ribbon, but if you are trying to insert an item and you don't see a dedicated tool for it, you can use the Component button. We use the Component button to bring in a variety of miscellaneous items. You can load components from the out of the box library like Plumbing Fixtures, Furniture, etc...you can create your own components or you can search for components online. So any time you need another component in your file that is not readily available in the Ribbon, all you have to do is this:



Other Components



1 Click the Component button, go to the load family button, browse through your available library and load in the components that you would like to use.

3 Go to the Insert Tab and use the Autodesk Seek button.

2 Go to the Insert Tab, Load from Library Panel, and then use the Load Family button.

4 Browse through the internet and look for Revit components.



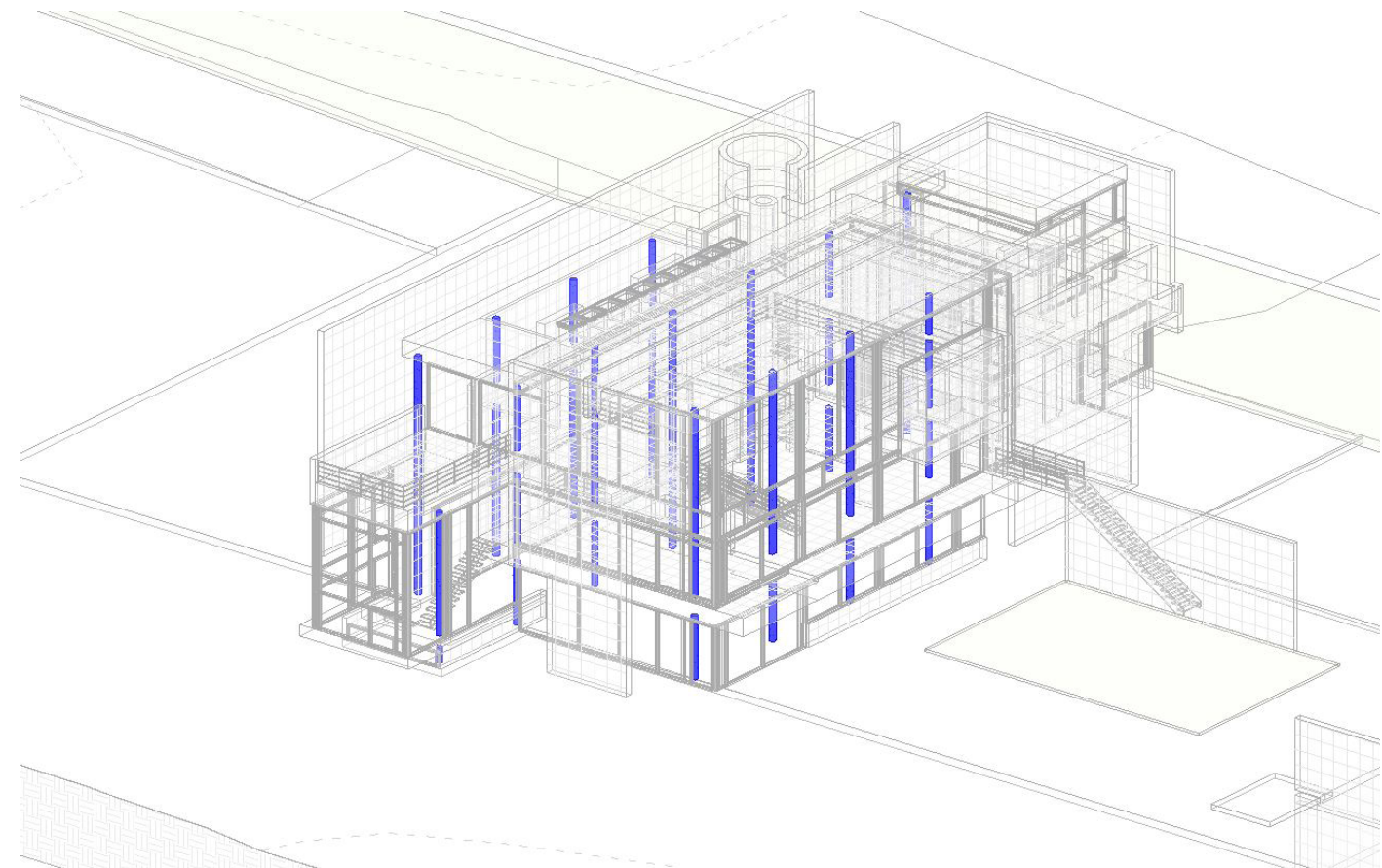
Basic Model Elements

Columns

Revit includes two types of columns: architectural columns, and structural columns. Typically, an **architectural column** is used to represent either a column wrap, or a rough placement column that will later be replaced by the **structural column**.

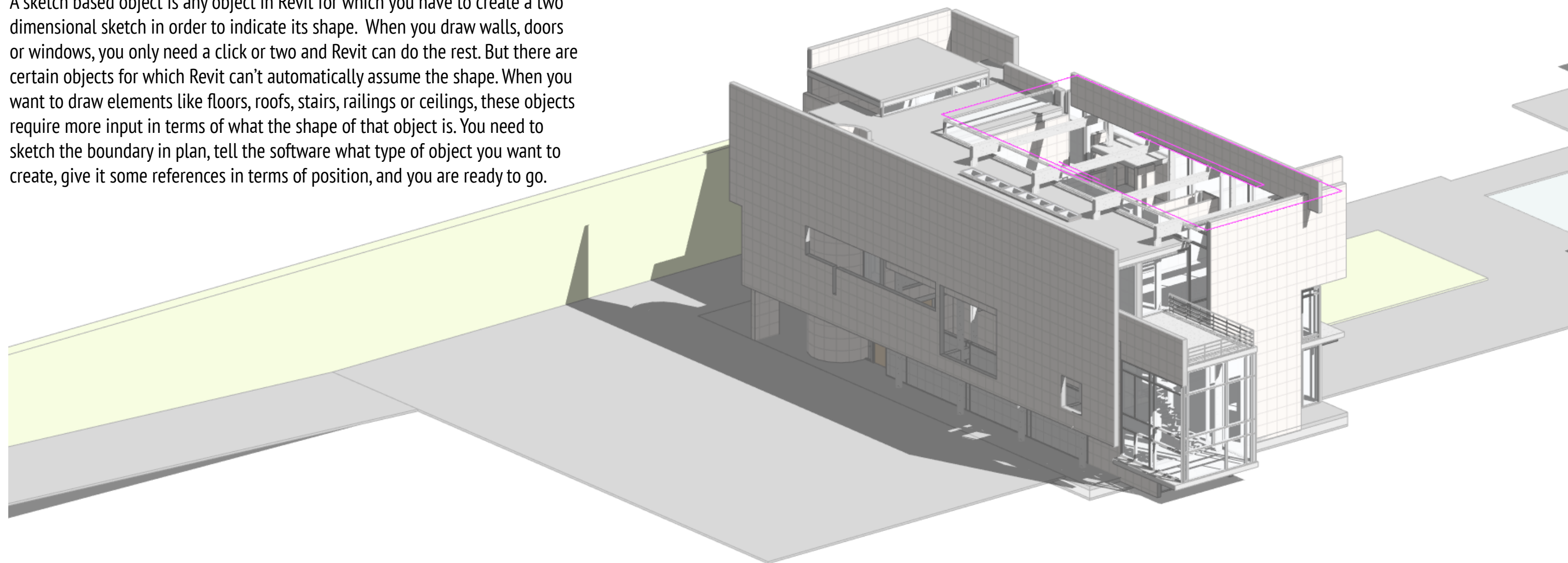
You can take advantage of the grids as the locations for your columns. You don't necessarily have to place your columns on grids, but if you place them at the **intersection** of two grids, if you later move the grid it will move the column along with it. One nice benefit of working with architectural columns is that they will automatically sense the presence of nearby walls and merge into the wall material.

One unique feature of structural columns is the multiple placement options on the **ribbon**. You can either place structural columns at the location of the architectural columns, or at the intersections of the grids. You just have to make a big selection and the software will detect either the architectural columns or the grids, and it will know where to place the structural columns.



Sketch-Based Modeling Components

A sketch based object is any object in Revit for which you have to create a two dimensional sketch in order to indicate its shape. When you draw walls, doors or windows, you only need a click or two and Revit can do the rest. But there are certain objects for which Revit can't automatically assume the shape. When you want to draw elements like floors, roofs, stairs, railings or ceilings, these objects require more input in terms of what the shape of that object is. You need to sketch the boundary in plan, tell the software what type of object you want to create, give it some references in terms of position, and you are ready to go.



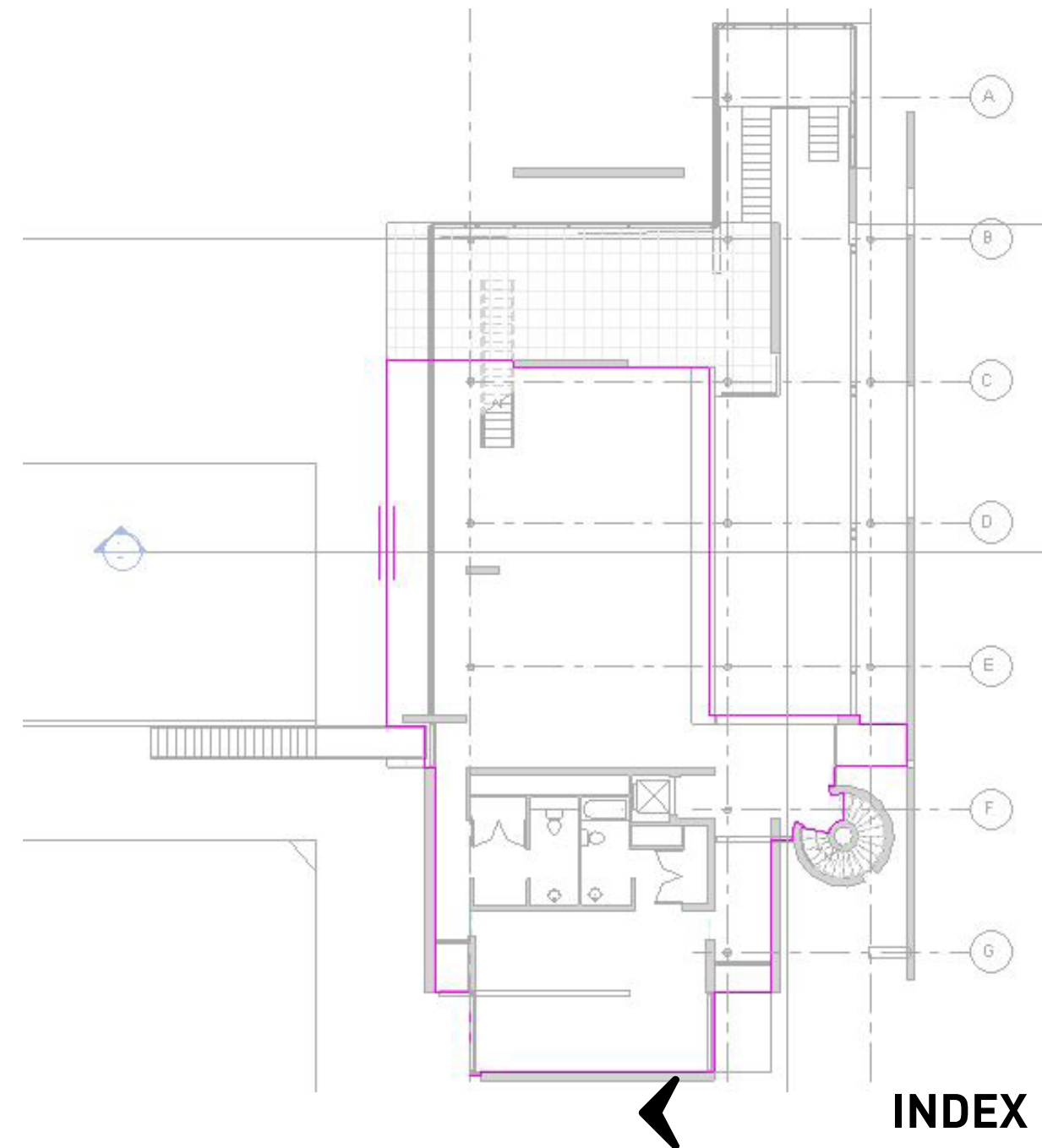
COMPONENTS

Sketch-Based Modeling Components

Floors

Floors are sketch based objects. To model a floor, you need to choose the **level** in which you want to place it, find the **Floor button** in the Ribbon, sketch its boundary in Plan, and choose the type of floor you want (or create a new type). You can also create **sloped floors** with the Slope Arrow Tool. If you have walls already modeled in your project, and you want those walls to define your floor boundary, you could use the **Pick Walls** button to pick those walls. If you do it this way you are indicating that these walls are the floor's boundary supports. By doing this, you will have the floor sitting on top of those walls and Revit will connect both elements properly in terms of **construction logic**. How can Revit know how to accurately connect the floor with the walls?

Remember that we mentioned that walls can be defined with **construction layers** and each layer can have a material, a thickness and a function. The construction layers can have different **Functions**: Structure, Substrate, Thermal/Air Layer, Finish 1, Finish 2, Membrane Layer...If both Wall components and Floor components have construction layers and each layer has a Function, Revit will know how these layers should interact with one another following a construction logic.



COMPONENTS

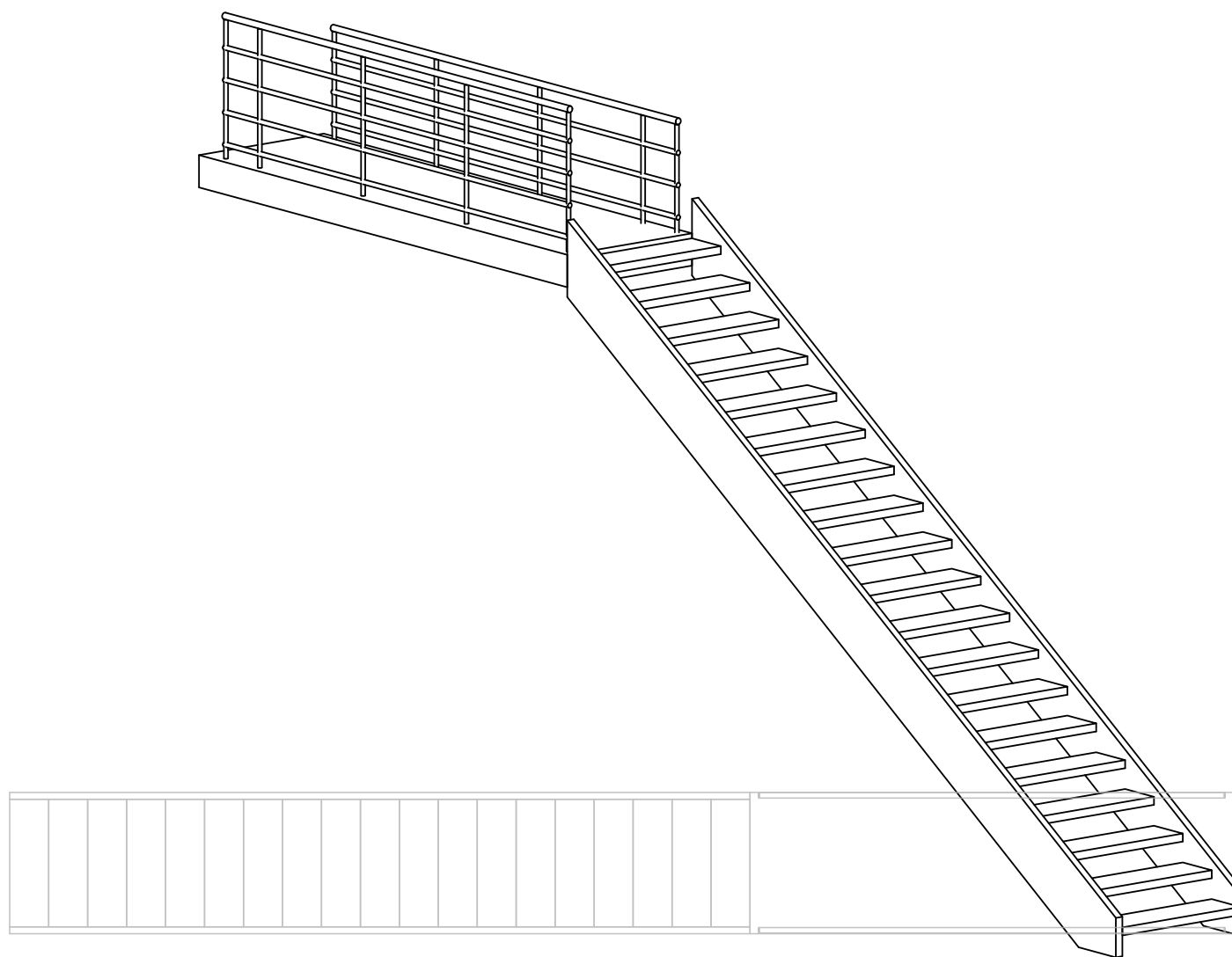
Sketch-Based Modeling Components

Stairs and Railings

There are two ways in which we can create stairs in Revit: Stairs by Sketch and Stairs by Component. Either way, to model a stair you should:

Tell Revit from what Level to what Level you want that Stair. Choose the **type** of stair that you want to model. Do the necessary clicks in the Drawing Area to properly model the stair (you just need to indicate the length of the runs and Revit will model the landings for you). It is important to understand that when you choose a certain Type of stair, you are specifying the Maximum Riser **Height** and the Minimum **Tread**, among other things, and with this information, Revit can calculate how many **risers** that stair needs to go from one Level to the other.

When you model a Stair, Revit will place the **Stair Railings** for you. You can choose what type of railings you want. You can also place railings on floors, if you need handrails. Railings are also sketch based components.



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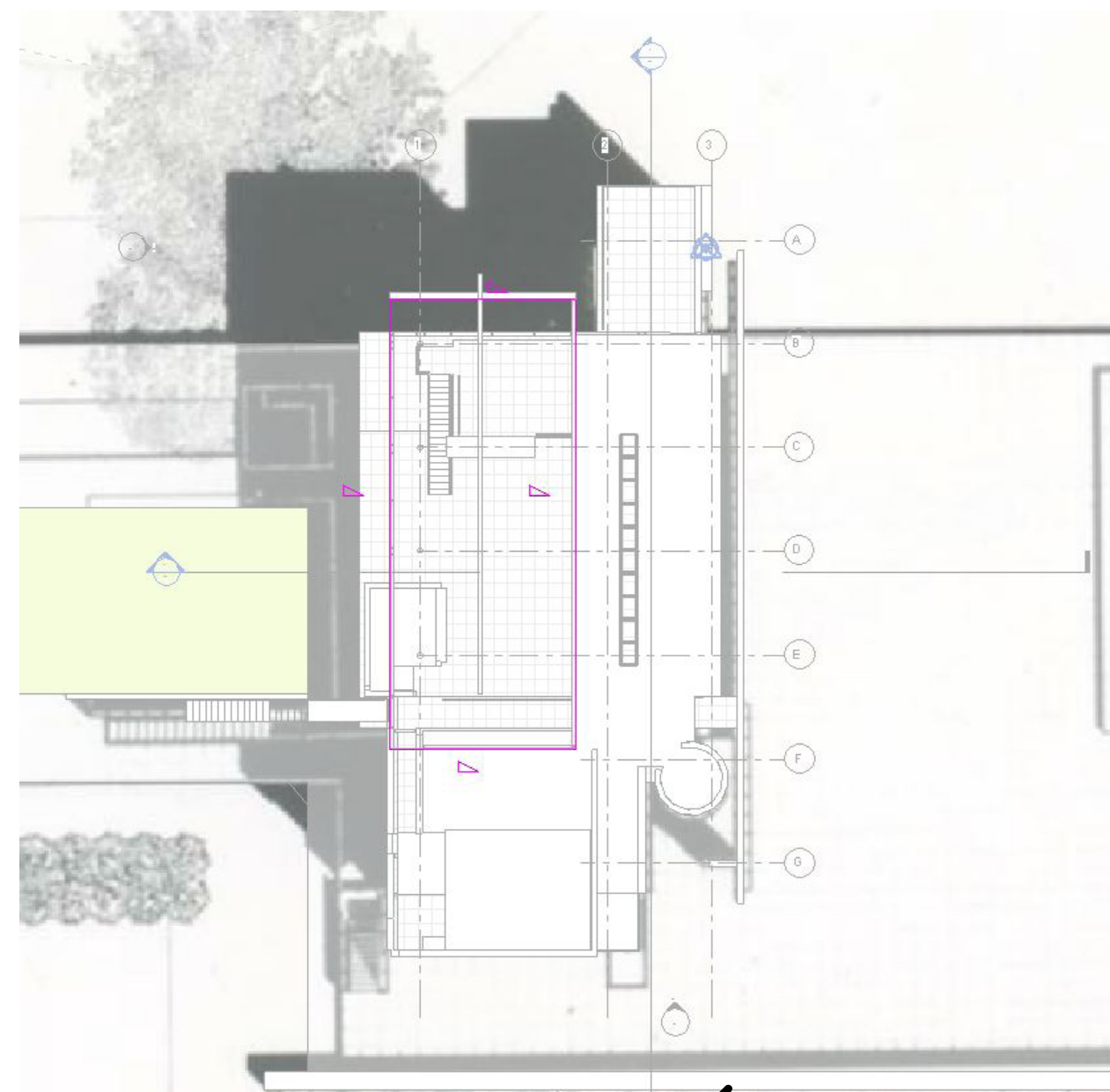
COMPONENTS

Sketch-Based Modeling Components

Roofs

There are two ways in which you can create roofs: Roof by **Footprint** and Roof by **Extrusion** (you can also create Roof by Face if you are working with masses). Roofs are also sketch based components. Footprint roof is very similar to a Floor except that typically you are going to apply a **slope** to it. But there are shapes of roofs that are not easily achieved with a Footprint roof. For those cases you can use the tool Roof by Extrusion, which is also a sketch based roof, but it is sketched in a **vertical plane** rather than a horizontal plane. As with Floors, all you sketch is the overall profile of the roof. How do we connect the Wall Geometry to the Roof Geometry?

All you have to do is select the wall or walls and use a **button on the Ribbon** to attach its top or base. This way you can attach walls with roofs. You can also use this tool to attach walls to floors or ceilings. When you attach a component to another component this connection will be maintained, even if the position of the component to which the other was attached changes.



INDEX



COMPONENTS

Sketch-Based Modeling Components

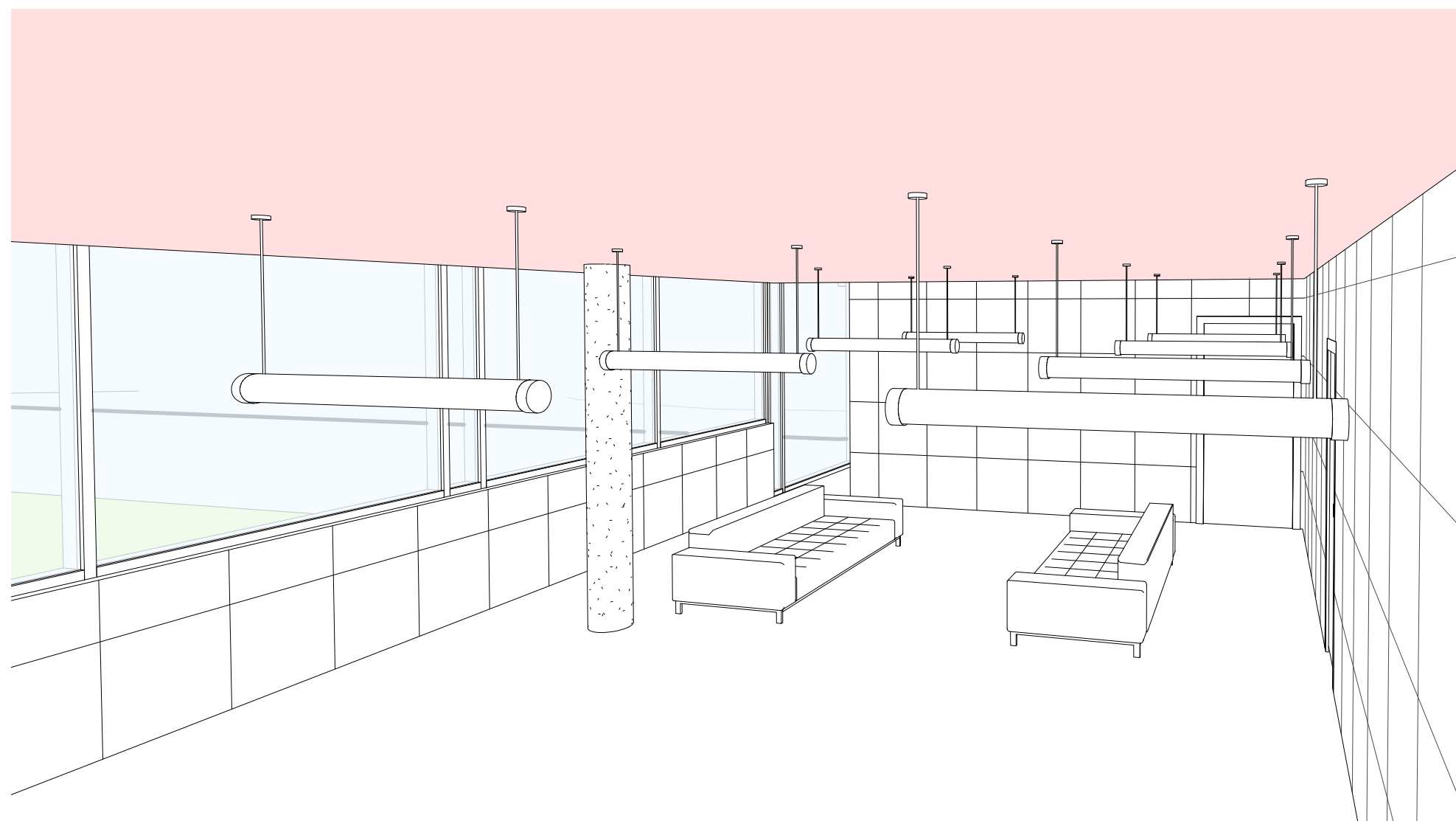
Ceilings and Lights

Ceiling elements are similar to floors. They are **horizontal planes** set at a certain height above the finished floor. They can include layers of material in their construction.

You can also create your own types.

The fastest way to create a ceiling is to use the **Auto-Ceiling** option. This tool automatically creates a ceiling from the walls that enclose a space. In cases where you don't have walls or where the **ceiling shape** is irregular you can sketch the ceiling.

It is very easy to add light fixtures to a ceiling. If there are no lighting components loaded in your project, then you just simply use the **component tool** and load the families you want. A light fixture is a ceiling hosted fixture. To place an instance you should click on a ceiling and the lighting fixture will be correctly placed in the ceiling at the position you want.



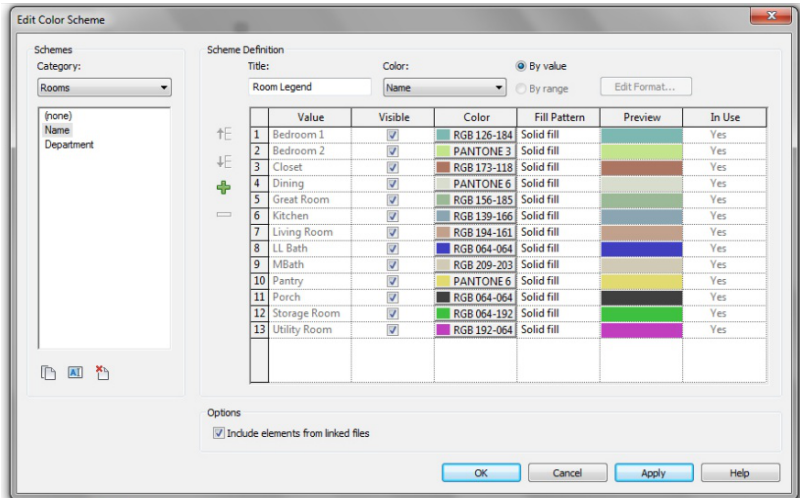
INDEX



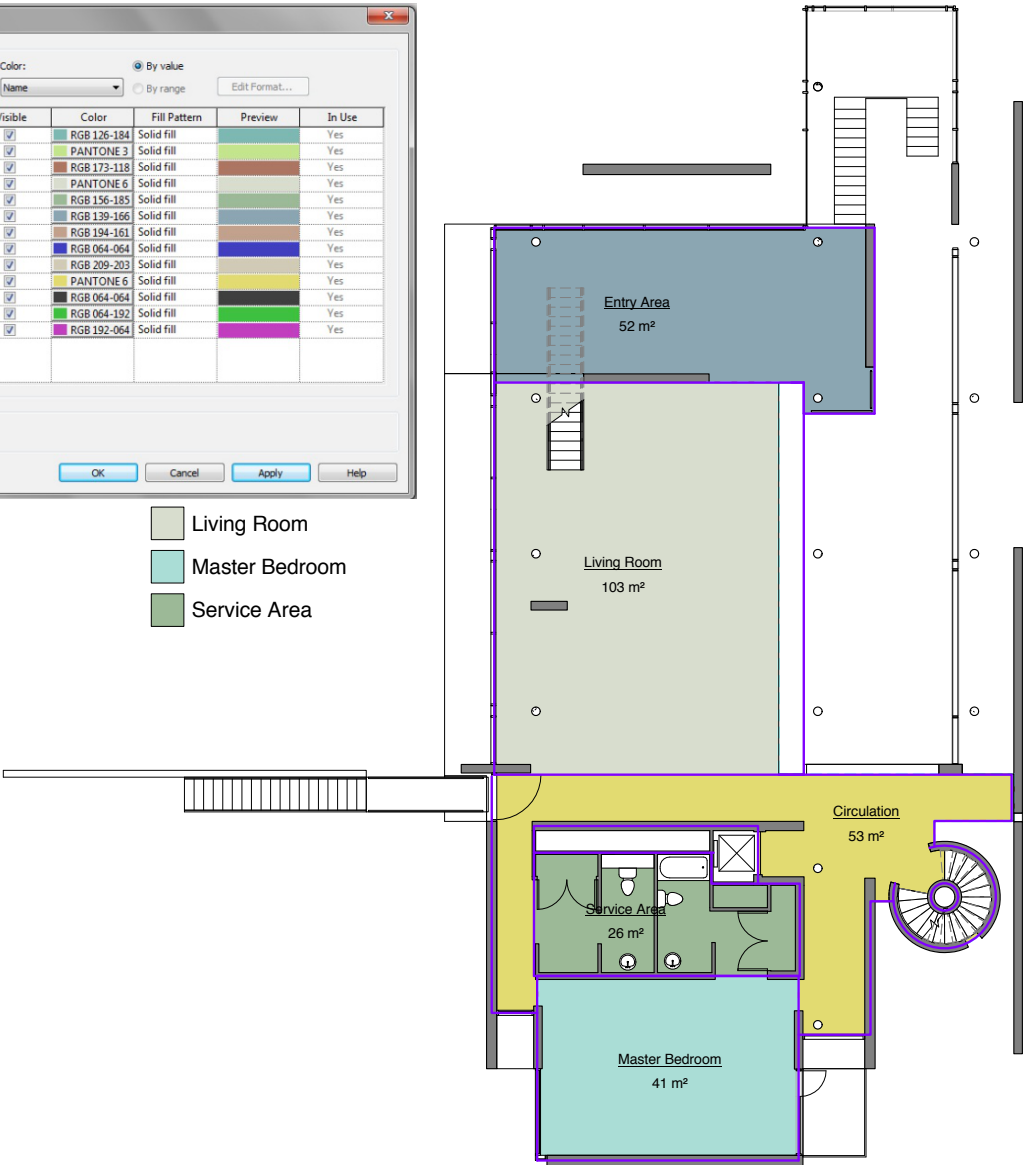
Adding rooms

In Revit, room elements are special objects that automatically conform to the shape of surrounding geometry and acquire the properties of that space. This is very useful, because if the walls that are defining that space change, the **room properties** (area, perimeter, etc...) will automatically update to the new situation. No manual updates are required. If you want to define a Room that is not completely enclosed by walls, you can use **Room Separator** lines, which are lines that also create room boundaries. The room object sees not only the walls but it sees those room separators as well.

Rooms come with **Room Tags** that show the properties of rooms. You can create different Room Tag Families to show different Room properties. Once you have placed Rooms in your project, you can easily apply **Color Schemes** based on any of the Room properties. You can also place Color Fill Leyends to explain those Color Shemes with just a simple click.



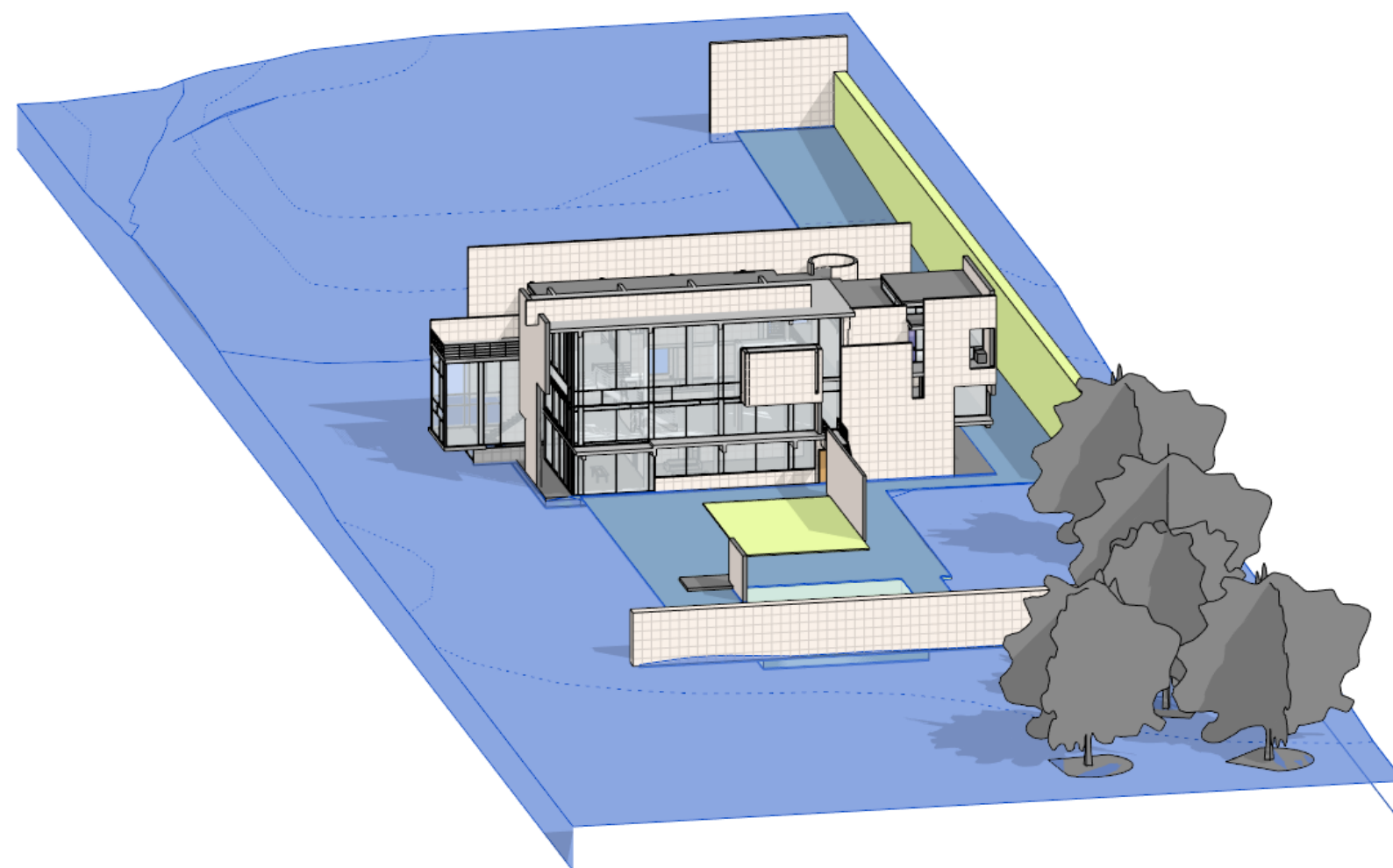
- Living Room
- Master Bedroom
- Service Area



The Site

There are different ways in which you can model a toposurface. You could place points at different heights and Revit would start connecting those points and building a **Topographic surface** from there, or you could create a toposurface from an imported instance, from example a DWG import instance. Once you have your toposurface you can change its material, Split it, create Subregions, change the Section cut material, etc... You could place trees and other **planting components** on top of that toposurface. If there were no Planting Components loaded in your Project, you could use the **Load Family button** which would open (by default) one of the Out-of-the-Box Libraries.

You should go to the Planting folder and choose the Planting families you needed. When you place these planting components in your project, they will automatically detect the toposurface and they will sit on top of it. In the **Properties Palette**, you can control their Offset. Notice that the names of the Planting components that you can find in the out of the box library, start with the letters **RPC**. Whenever you see the letters RPC with anything related to a component in Revit, or other rendering programs, it indicates that when that object is rendered, it is going to look like the real object. An RPC is actually made of a set of real images of real things (trees, people, cars...) so when it is rendered, it will look real.



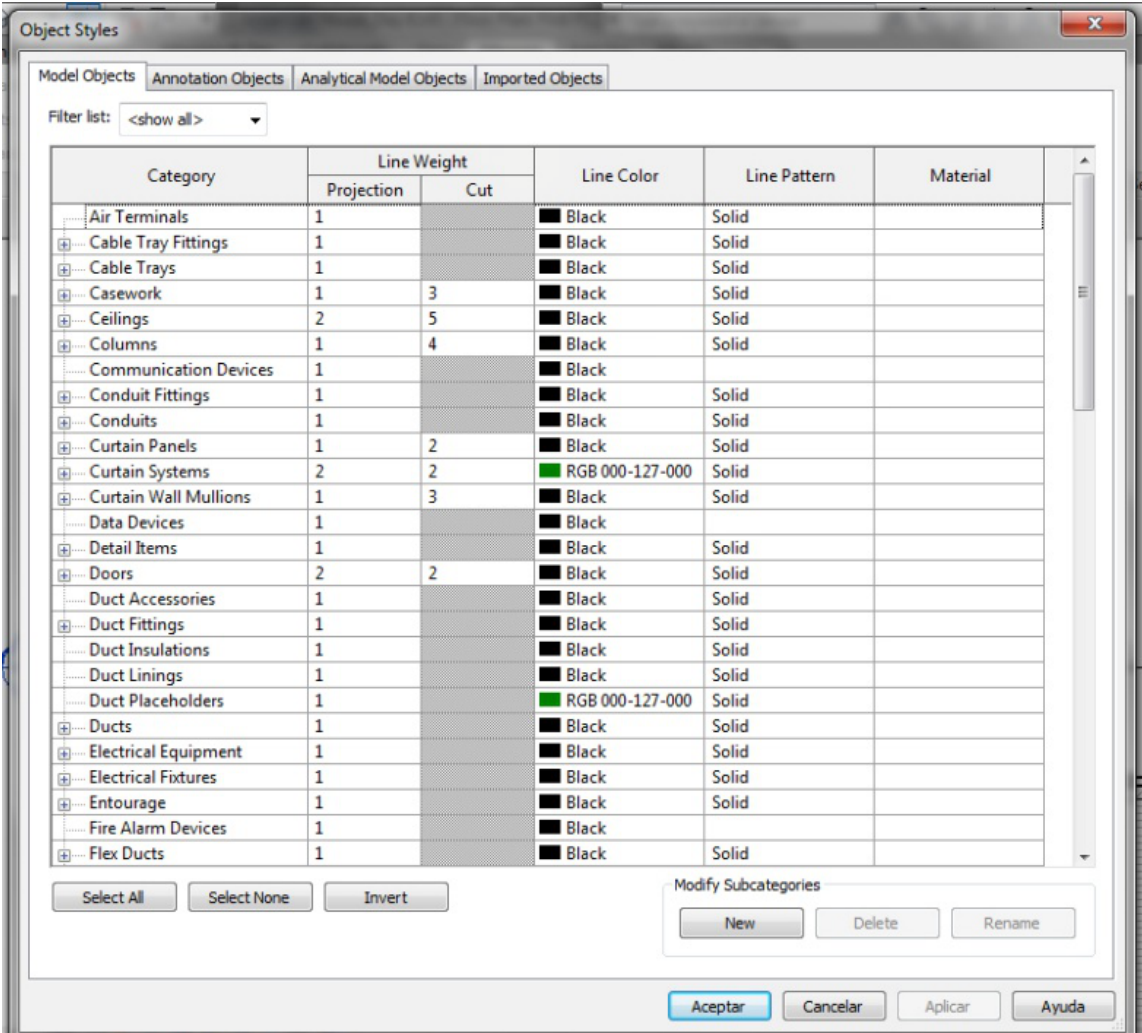
Visibility and Graphic Controls

Using objects styles

Object styles are the graphic styles of the different objects in Revit. Let’s study this concept in comparison to AutoCAD. In AutoCAD every line belongs to a layer, each layer has a color assigned to it and each color has a plot style assigned to it. We create the layers we need and we create a plot styles file for those layers. For every line we draw, we should decide what layer it belongs to.

In Revit, we don’t use layers. Elements in Revit are organized in **Categories** (Doors, Windows, Walls, Floors, Roofs...) and each Category has its own “Object Style”. This means that every object that belongs to a certain Category will have the “Object Style” assigned to that category. An **object Style** is defined by a Line Weight for Projection, a Line Weight for Cut, a Line Color, a Line Pattern and a Material. These values are defined in the Object Styles dialog for each Category, but you can change these values if you want. Since it is not the same to see a Floor Plan in a 1/100 scale than to see it in a 1/5 scale, the values assigned in the Object Styles dialog have different Line Weights assigned for different drawing scales. We can see this in the **Line Weights Dialog**.

Of course, we could also change these Line Weights if we wanted to. So each Category has an Object Style that defines the way we will see every object that belongs to that category, and this Object Style will adjust to the scale of the drawing accordingly. Most of the time the default values work fine while learning the essentials and we won’t need to change them. So, we basically have to worry about **Line Properties** and **Materials** no more.

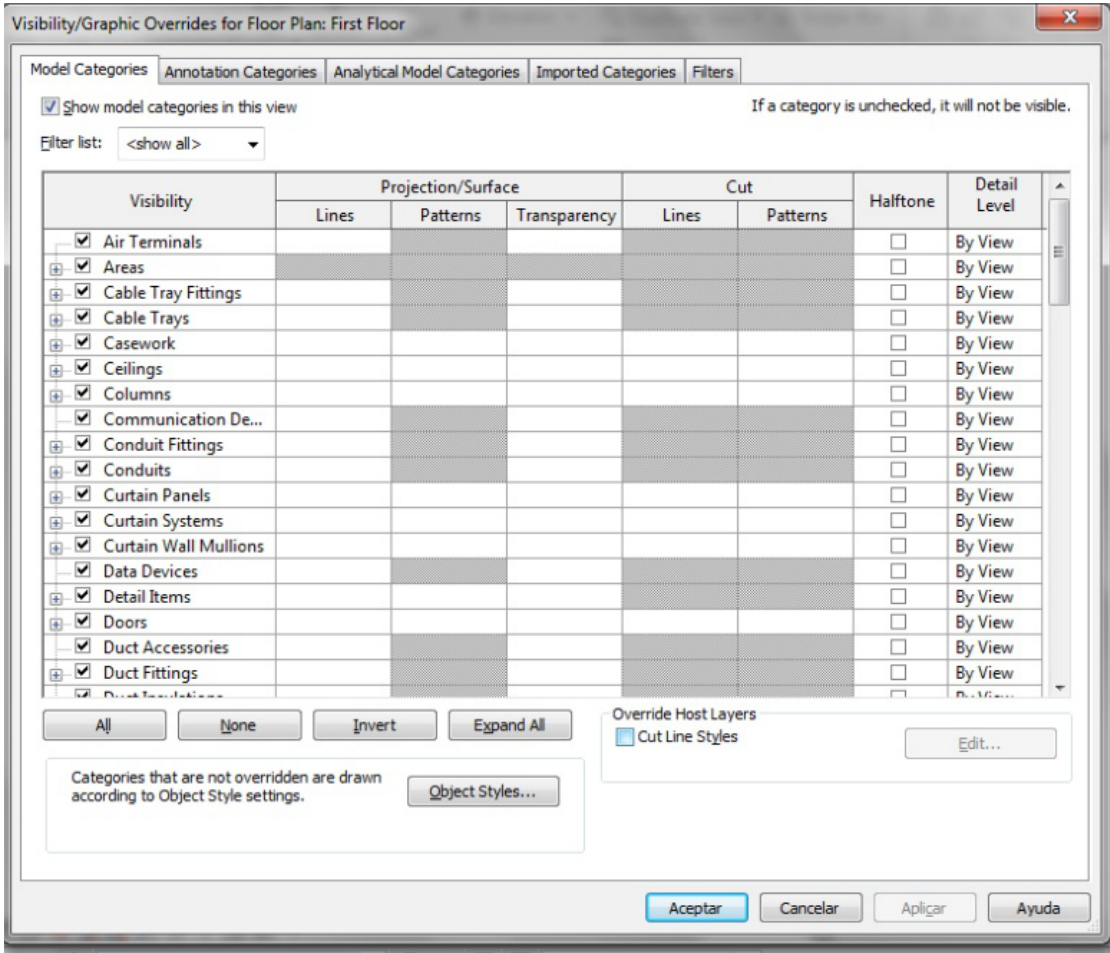


Visibility and Graphic Controls

Working with visibility and graphic overrides (VG)

So what happens if you want to change the way an object is displayed just for one particular view. For example, you want the Furniture Category to be seen with a pink color in projection just in one **Floor Plan View**, but you want to see it with the “Object Style” defined for that Category in the rest of the Views. Or what happens if we don’t want to see the Furniture Category in one **particular Section View**, but we want to see it in all the rest of the Views. All this can be managed per View with the **Visibility/Graphics Override dialog**. You can override the Object Styles for any particular view, to see that view exactly the way you want, by overriding the Object Styles defined in the project. You can also decide, for each view, which categories you want to see and which ones you don’t want to see.

By default, the Visibility/Graphics Override dialog has most of the categories visible and with the Object Styles applied to each category, but you can override the Object Styles for any Category in each view, or decide which categories you want to see and which ones to don’t want to see in each view. So remember, the Visibility/Graphic Overrides is **View Specific**, you can change the settings just for a specific view.



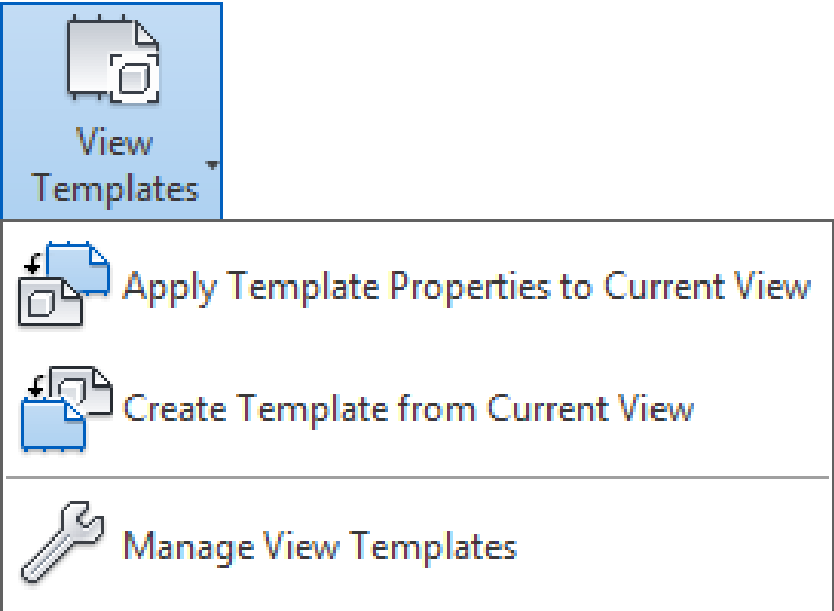
Visibility and Graphic Controls

You can use View Templates to capture all of the settings that you've applied to a view and apply those settings to other similar views. You can create your own custom View Templates. You can do this in two different ways.

1 Apply Template Properties to Current View.
By choosing this option we are copying the settings from the **View Template**, but we are still able to change any setting of that view.

2 Apply the settings from the View Template as a **Property** of that view, so that the View is controlled by that View Template. If you needed to change some Property of the view you would have to change it in the View Template.

Using View Templates

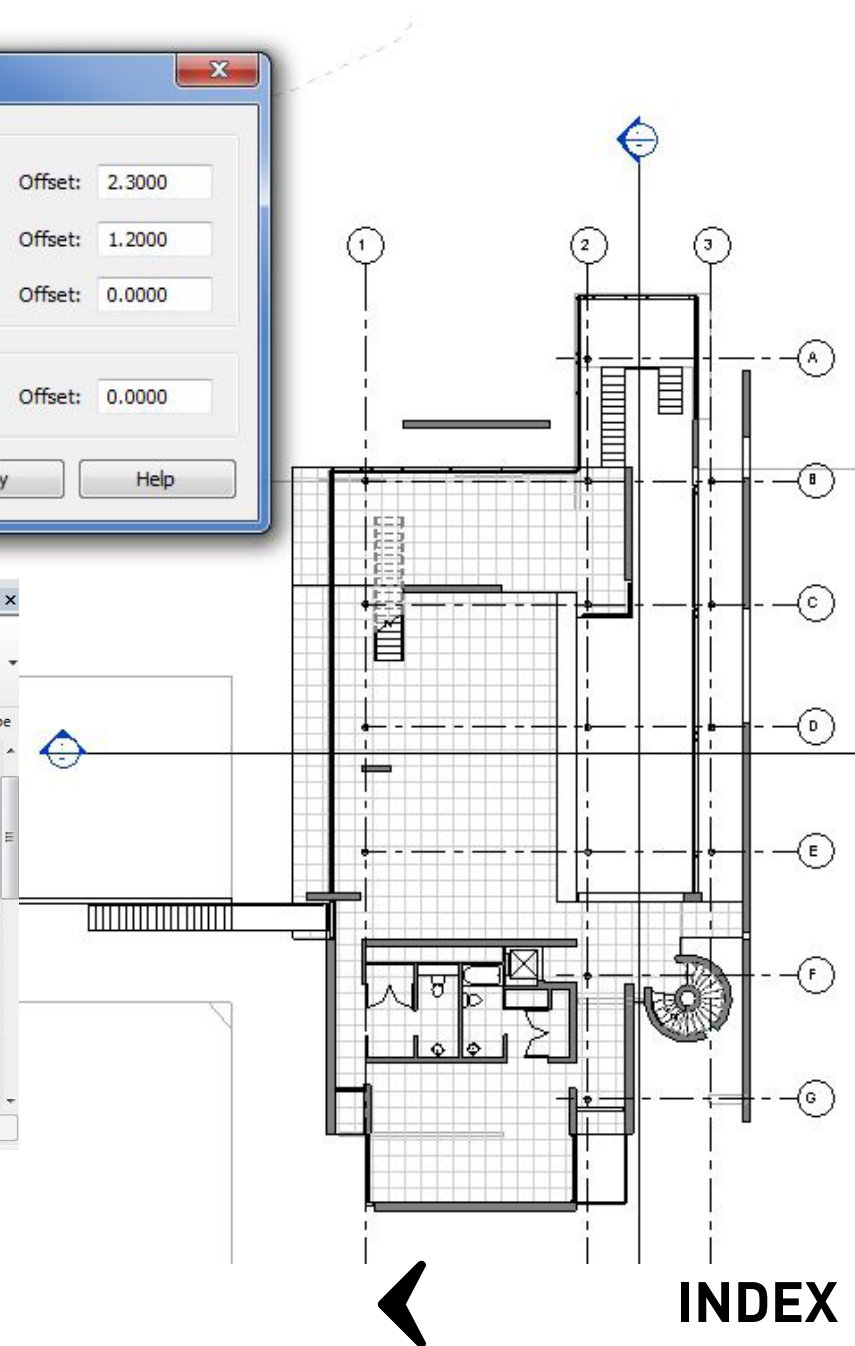
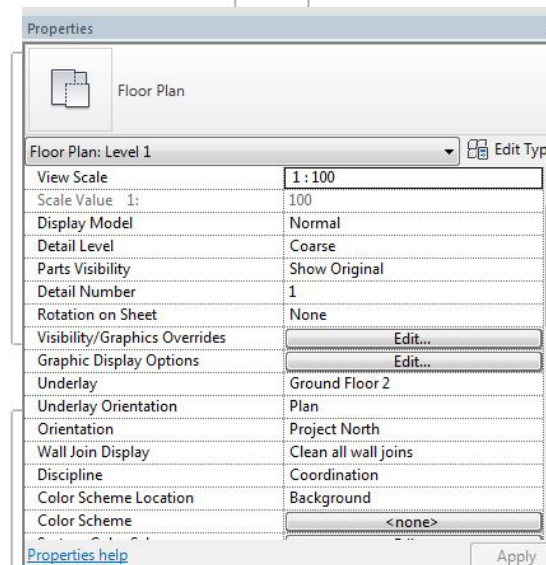
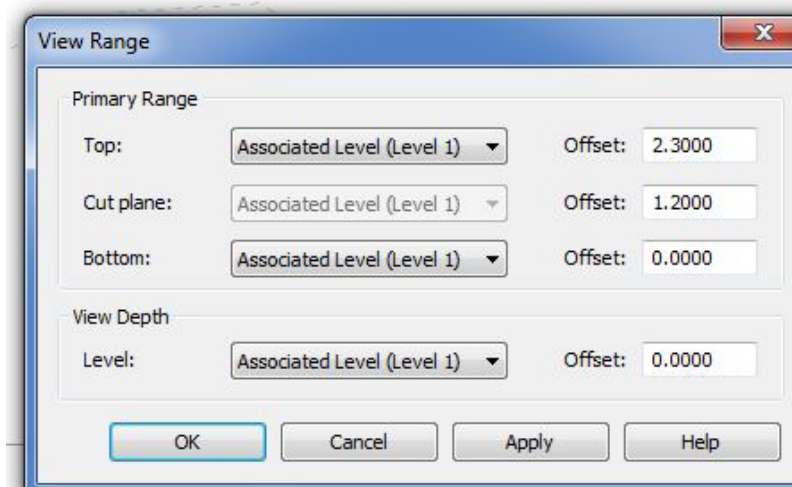


Visibility and Graphic Controls

Understanding View Range

A Plan View is basically a horizontal section. We are Cutting the building at a predetermined height, and looking down into the model. **View Range** is one of the Properties of a View, and we can find this property in the Properties Palette. You will see four settings in the View Range Dialog. There are three settings for the **Primary Range**, and one setting for the **View Depth**. These settings have default values. For the Primary Range, there is a Cut plane set at a certain height, and a Bottom and Top plane. For the View Depth there is a plane set at a certain height. **Cut Plane** determines where your floor plans are being cut from. Bottom Plane indicates the Level/Plane that you will see in projection.

You will be able to see certain categories that are located between the Cut Plane and the Top Plane. These Categories will be shown with the <Overhead> Line Style. Every object that is between the Bottom Plane and the View Depth Level plane will be shown with the <Beyond> Line Style. You can determine the height of all of these planes to see the view as you want.



Schedules, Legends and Tags.

Schedule Views

With schedules we focus on the Information part of Building Information Modeling. Schedules are an important part of any architectural document set. What makes schedules unique in Revit, is that they are **live Views** of the model. They are not just static reports that you have to draft out line by line. In Revit, when you generate a schedule it queries the **model** and populates the schedule based on the objects that are already in the model. If you have a really large project, and you can't locate a particular element, you can also use your schedules to help you locate elements in your model.

There are many formatting options available to make a schedule more eligible and to ensure its usefulness. By default, a schedule will report the **items** in the order that they were added to the model but you can start to sort the schedule in some more logical ways to make your information more usable. In the **Schedule Properties** dialog you have different Tabs that allow you to Filter, Sort, Group, Format or change the appearance of the schedule. Using these tools you can make your schedules more eligible and useful.

- 1

Go to the appropriate tool to create a new Schedule.
- 3

Choose a name for the Schedule (Revit will suggest the name, but you can change it).

- 2

Select a Category.
- 4

In the Schedule Properties dialogue, select at least one field to include in your schedule. The available fields are the list of properties available for the category that you selected. The fields will become the columns in the Schedule view.

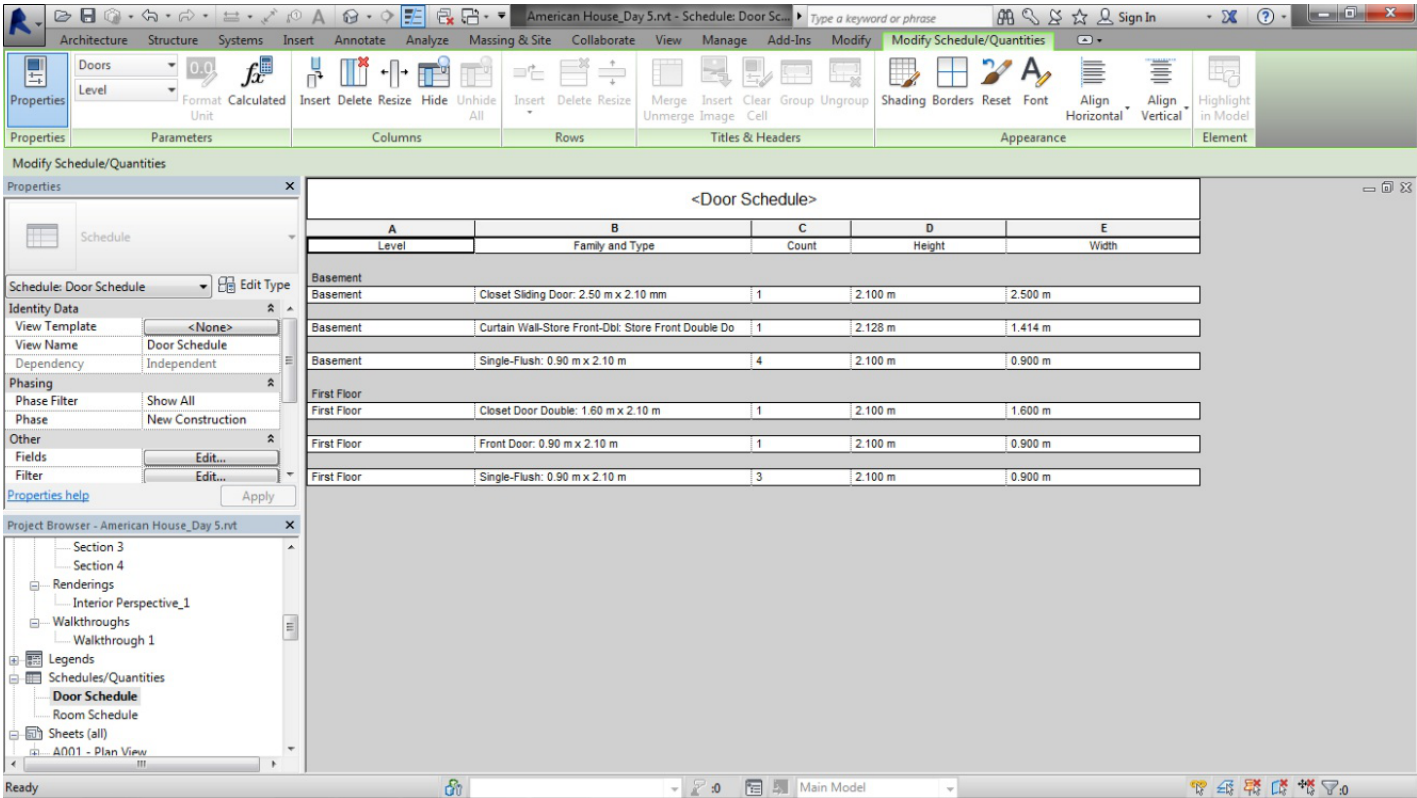
<Door Schedule>				
A	B	C	D	E
Level	Family and Type	Count	Height	Width
Basement				
Basement	Closet Sliding Door: 2.50 m x 2.10 mm	1	2.100 m	2.500 m
Basement	Curtain Wall-Store Front-Dbt: Store Front Double Do	1	2.128 m	1.414 m
Basement	Single-Flush: 0.90 m x 2.10 m	4	2.100 m	0.900 m
First Floor				
First Floor	Closet Door Double: 1.60 m x 2.10 m	1	2.100 m	1.600 m
First Floor	Front Door: 0.90 m x 2.10 m	1	2.100 m	0.900 m
First Floor	Single-Flush: 0.90 m x 2.10 m	3	2.100 m	0.900 m

Schedules, Legends and Tags.

Schedule Views

Since a schedule is a live view of the model, you can use it to select things or you can make changes directly in the schedule. Those changes will immediately be reflected back in the model. You basically use whichever view is convenient for you (graphic view or schedule view) to make an edit. This is just one more example of how everything is **interconnected** in Revit. If you have a really large project, and you can't locate a particular element, you can also use your schedule to help you locate elements in your model. There are many formatting options available to make a schedule more eligible and to ensure its usefulness.

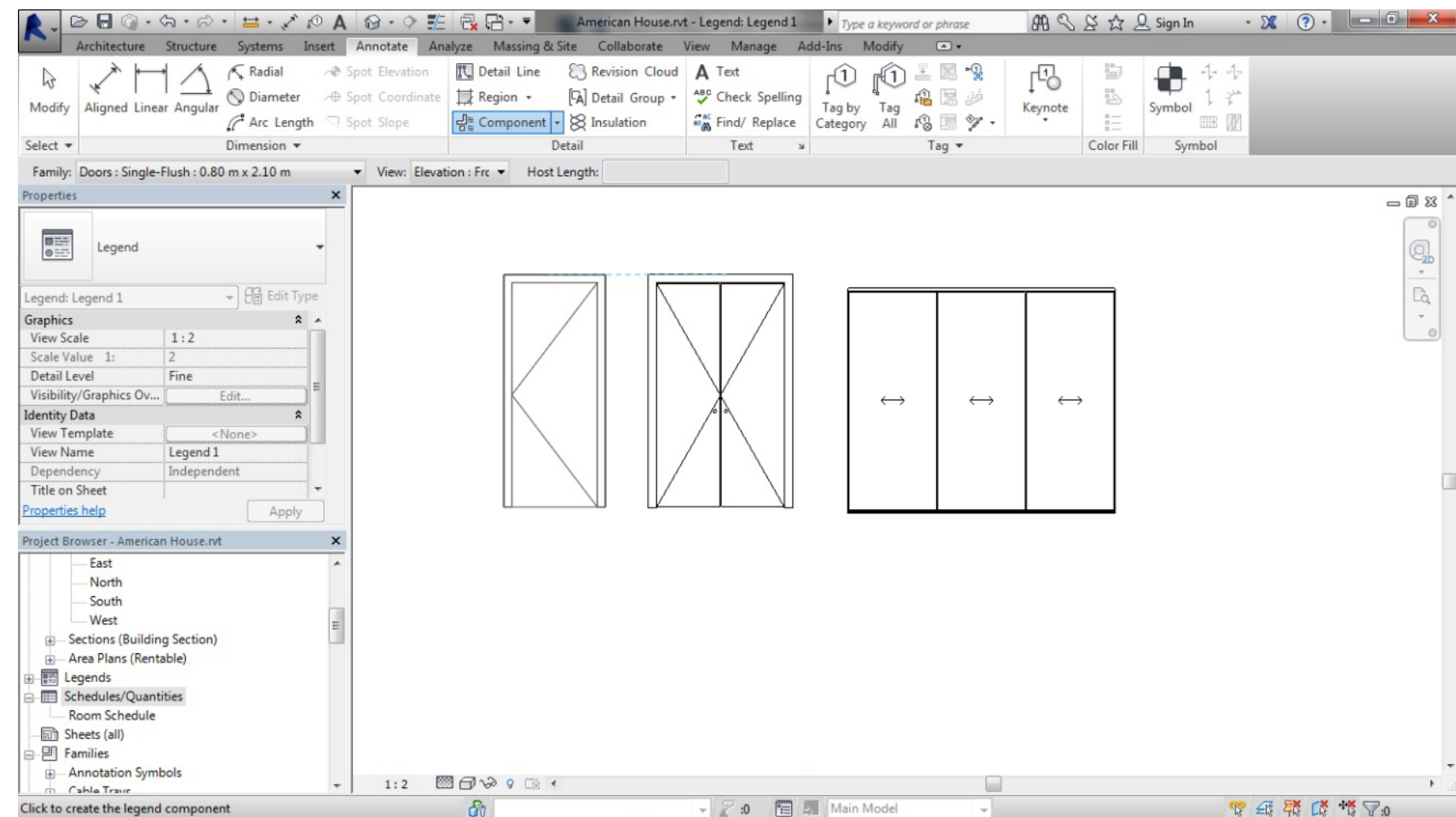
By default, a schedule will report the items in the order that they were added to the model but you can start to sort the schedule in some more logical ways to make your information more usable. In the Schedule Properties dialog you have different **Tabs** that allow you to Filter, Sort, Group, Format or change the appearance of the schedule. Using these tools you can make your schedules more eligible and useful.



Schedules, Legends and Tags.

Generating a Legend

If you would like to add a legend to your project, showing each kind of door or window, for example, there's no need to draw this manually. A **Legend view** in Revit allows you to place a graphical representation of any family used in your project. You'll be able to control the scale and the level of detail. And you'll be able to insert any family as either a plan or an **elevation symbol** and add the notes and formatting you consider appropriate.

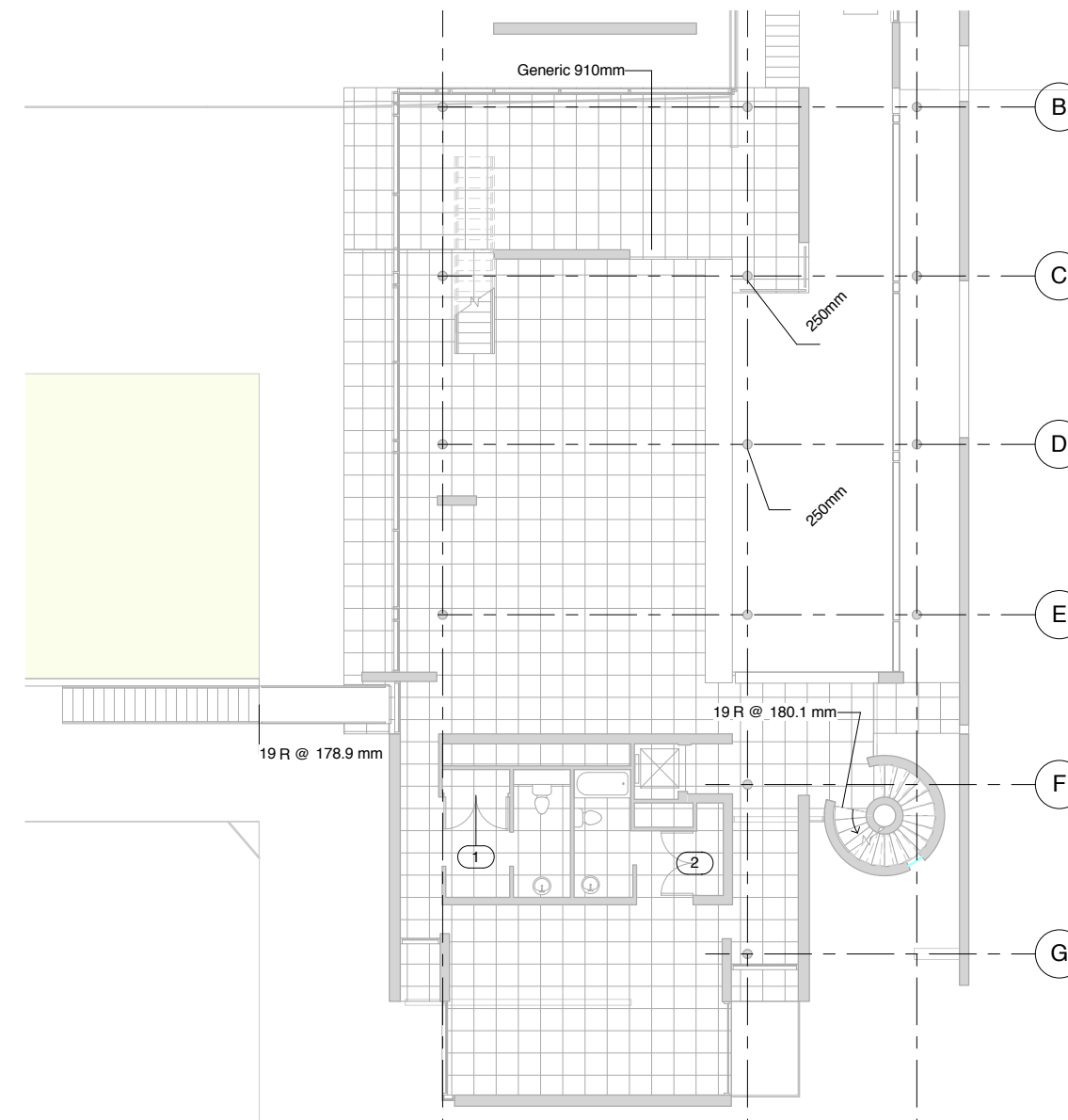


Schedules, Legends and Tags.

Tags are very common in architectural drawings. We typically add tags to rooms, doors, windows, walls, and many other items. In Revit, the element, the tag and the schedule are all linked together. The tag basically asks a question to the object to which it is attached and reports the result onscreen. It is asking a question like: "What is the value of this Property for this object" and it gives the answer as a text or a number on screen. In the Ribbon there is the **Tag by Category** command and the **Tag All Not Tagged** command. With the Tag by Category command, Revit will see the category of the object under your cursor and it will choose from a list the appropriate tag. You can load different **Tag Families** in your project. Remember that it is not the Tag that holds the value but the tag is just reporting a value after looking at a property of a particular object.

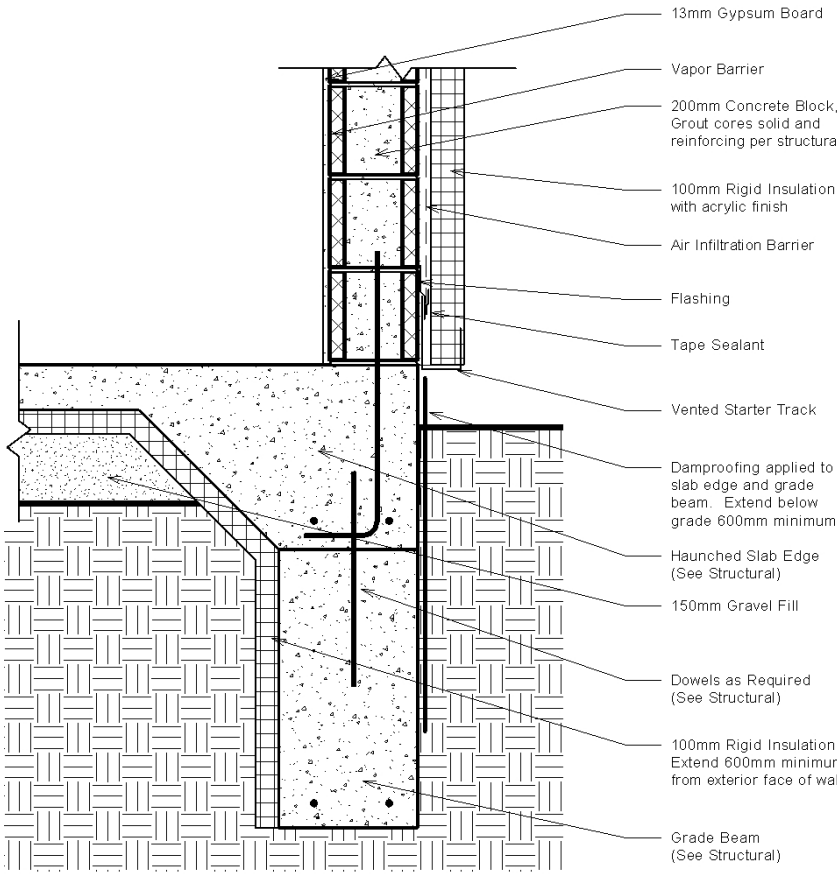
With the Tag All Not Tagged command you can tag every object of the category you choose that doesn't already have a tag. This way you don't have to Tag objects individually. So as you can see adding tags to a Revit view is quick and easy. You can either use the Tag by Category feature, which will allow us to select any object and tag it. Or you can use the Tag All Not Tagged feature which looks at the entire view and tags any object of a particular category that doesn't have a tag.

Understanding Tags



Annotation and Details

You can add text, dimensions, symbols, detail components and filled and masking regions to you drawings. There comes a **point** in your projects where your model is at a level of completion where you want to begin adding some **annotation**, to clarify design intent or to prepare the drawing for printing. You can very easily annotate your drawing. Annotations are **2D components** that appear only in the View in which they were placed. For example, the way you add dimensions in Revit is very simple and fast. You can add strings of dimensions quickly or add dimensions to an entire wall just with a simple click. Remember that not only do we have model families, we also have **annotation families**.



We can create **Views** like Plans, Sections, Elevations, etc...based on our model. Views in Revit are just “windows” looking at our model. **Elevation Views** are usually already created in Project Templates. You are just indicating that you want to look at your model from a particular point of view. **Section views** can be created with just two clicks. Since you are working with a 3D model, you will have the section automatically. You are just telling the software what vertical plane it needs to use to cut your model and show you a section. You can also create **callout views** (enlarged portions of any of your 2D views) easily.

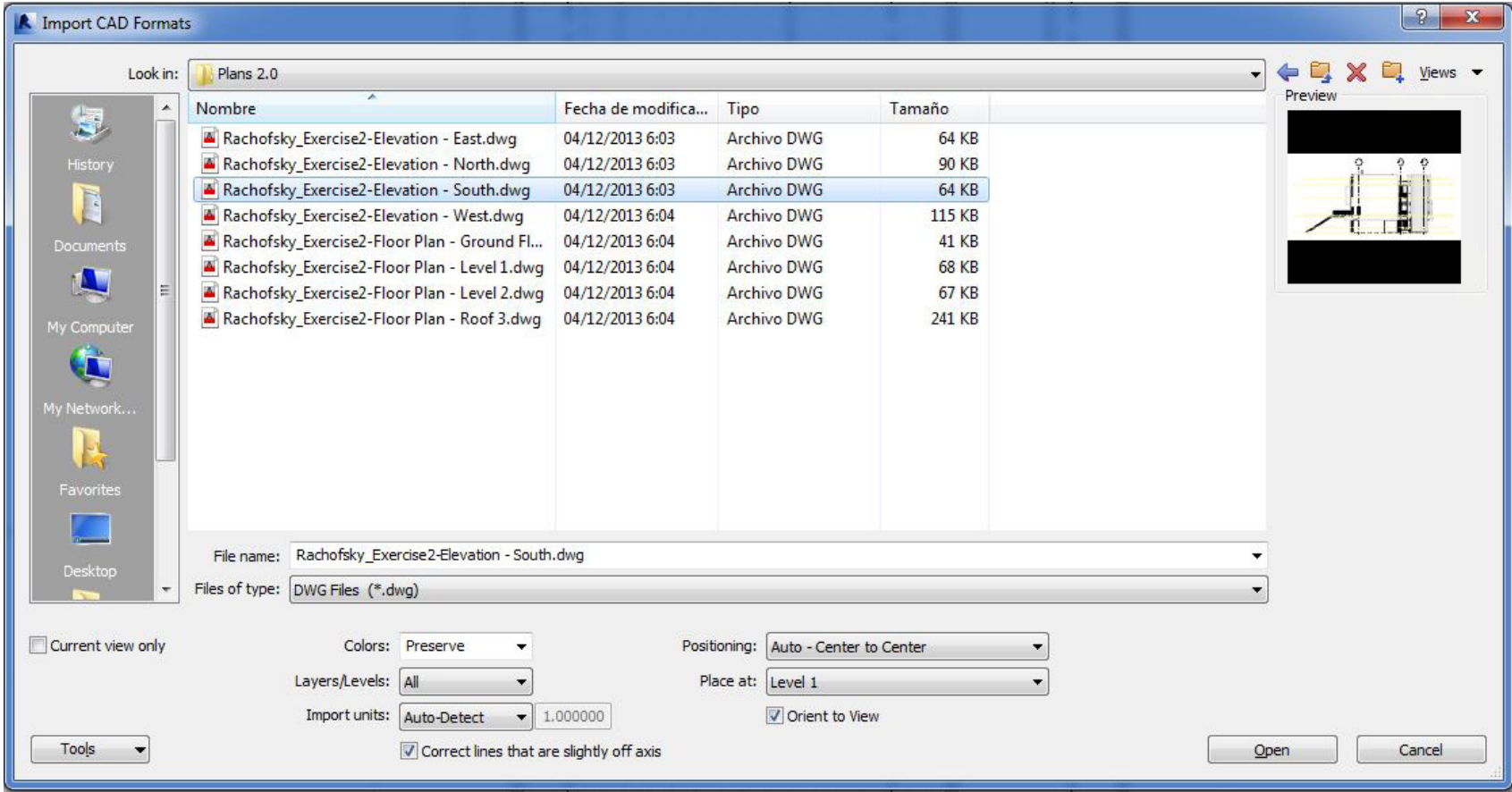
Creating 2D views



Links and Imports

Linking and Importing AutoCAD DWG files

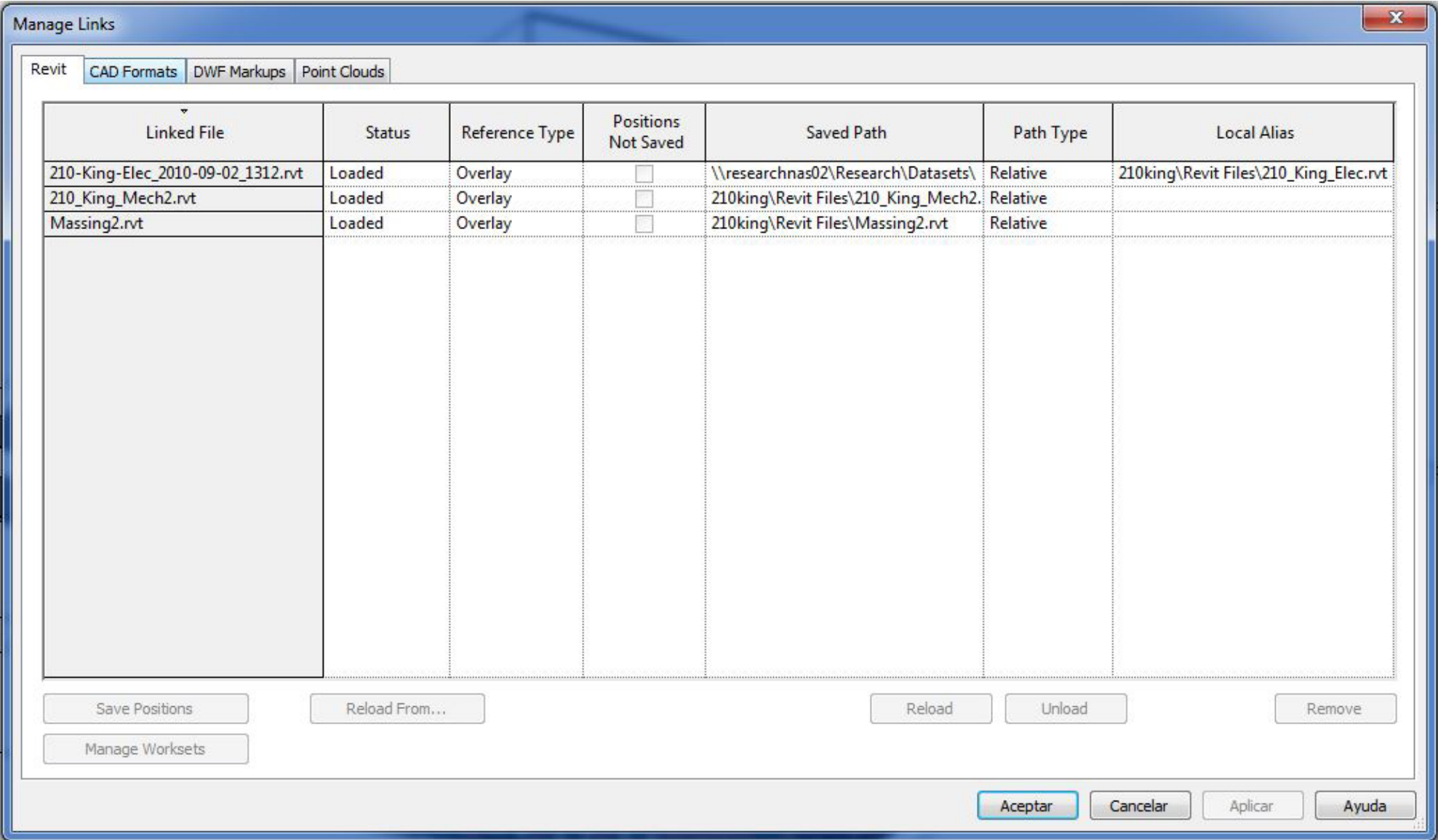
Revit supports the import and export to many popular **CAD formats**. It is quite likely that some of your projects will have the need for a **DWG** at some point. You can link in a DWG file or you can import a DWG file. When you link a DWG, you maintain the connection back to the original file so if the original file changes you can reload to capture the latest changes. If you import a DWG, you embed it into your project.



Links and Imports

Creating Revit Links

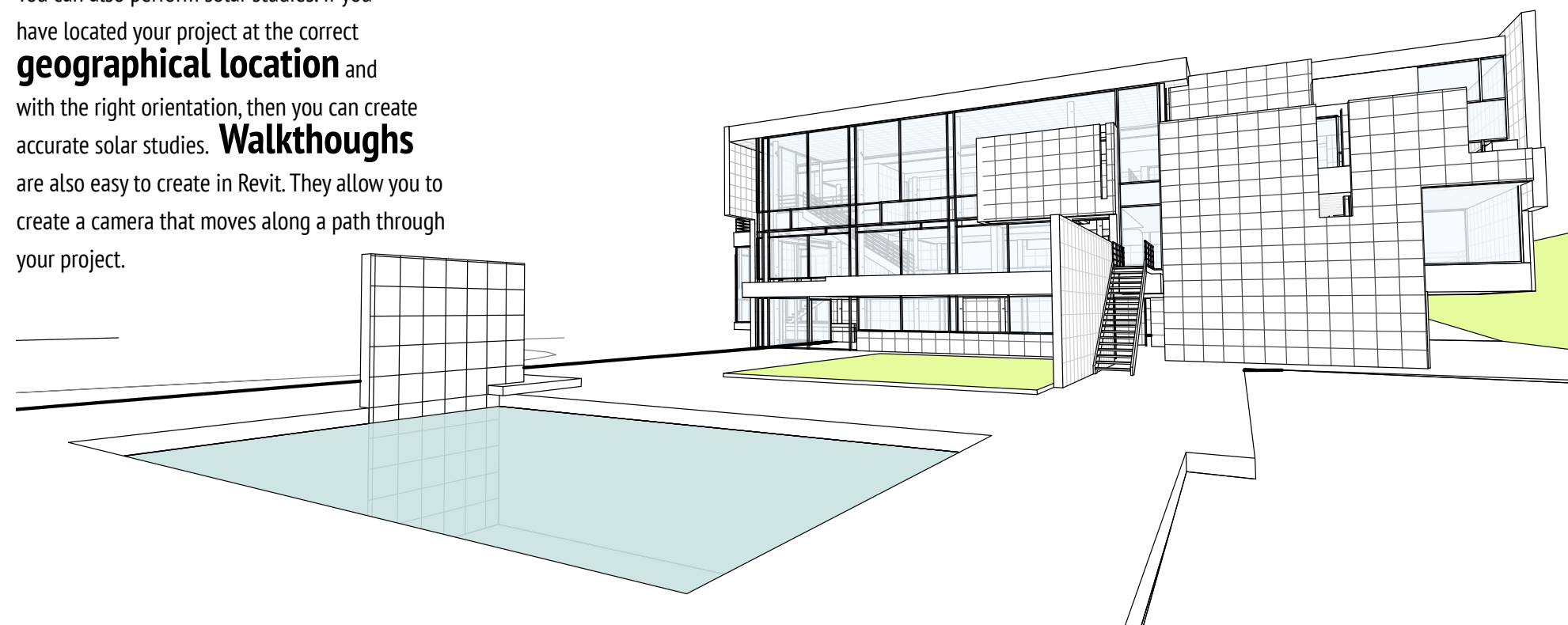
There are lots of **scenarios** where you might want to link one Revit project into another. Perhaps, you are working on a multi building project or maybe you have multiple disciplines (Mechanical, Electrical, Structural...) and you need to **link** all disciplines in the same model. Whatever the scenario might be, you can take one Revit project and link it into another Revit project. If the original Revit file changes, you can reload it to see the latest changes.



Presentations

You can easily create interior or exterior perspective views of your model just with a single click. You can then use a variety of **graphic display options** to make these views appealing. If you have created your model with enough level of development, and you have set the materials for at least the outer most construction layers of your components, it is very simple to get rendered images your of your model. The general process to create a render is to first create a **3D view**, set up the lighting and materials needed, configure the settings (resolution, lighting scheme, background settings, exposure), and finally click the **Render button**. If you want to save renderings then you can either save them to the project or export them out to files.

You can also perform solar studies. If you have located your project at the correct **geographical location** and with the right orientation, then you can create accurate solar studies. **Walkthroughs** are also easy to create in Revit. They allow you to create a camera that moves along a path through your project.

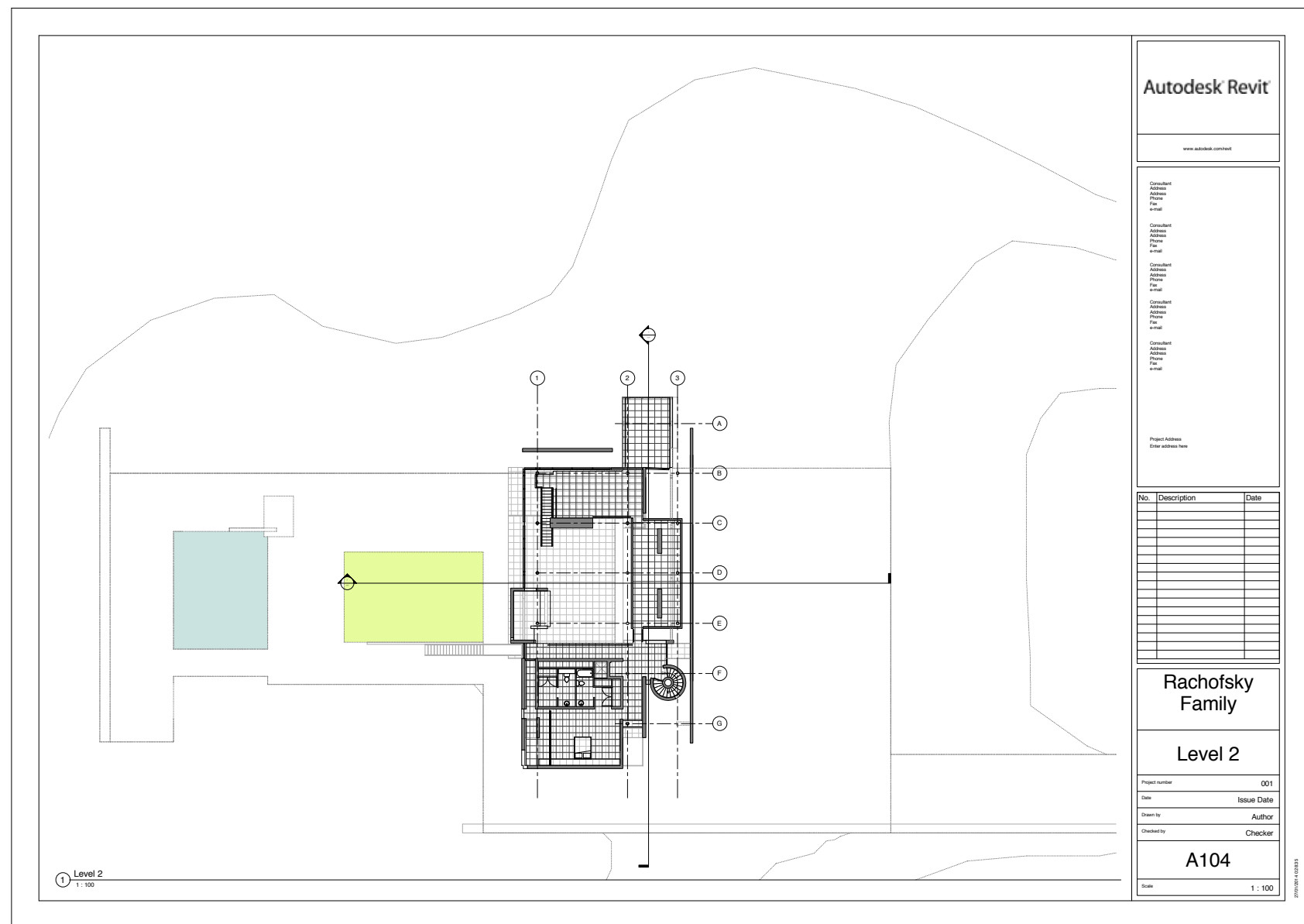


Architectural

Sheets, Plotting and Publishing

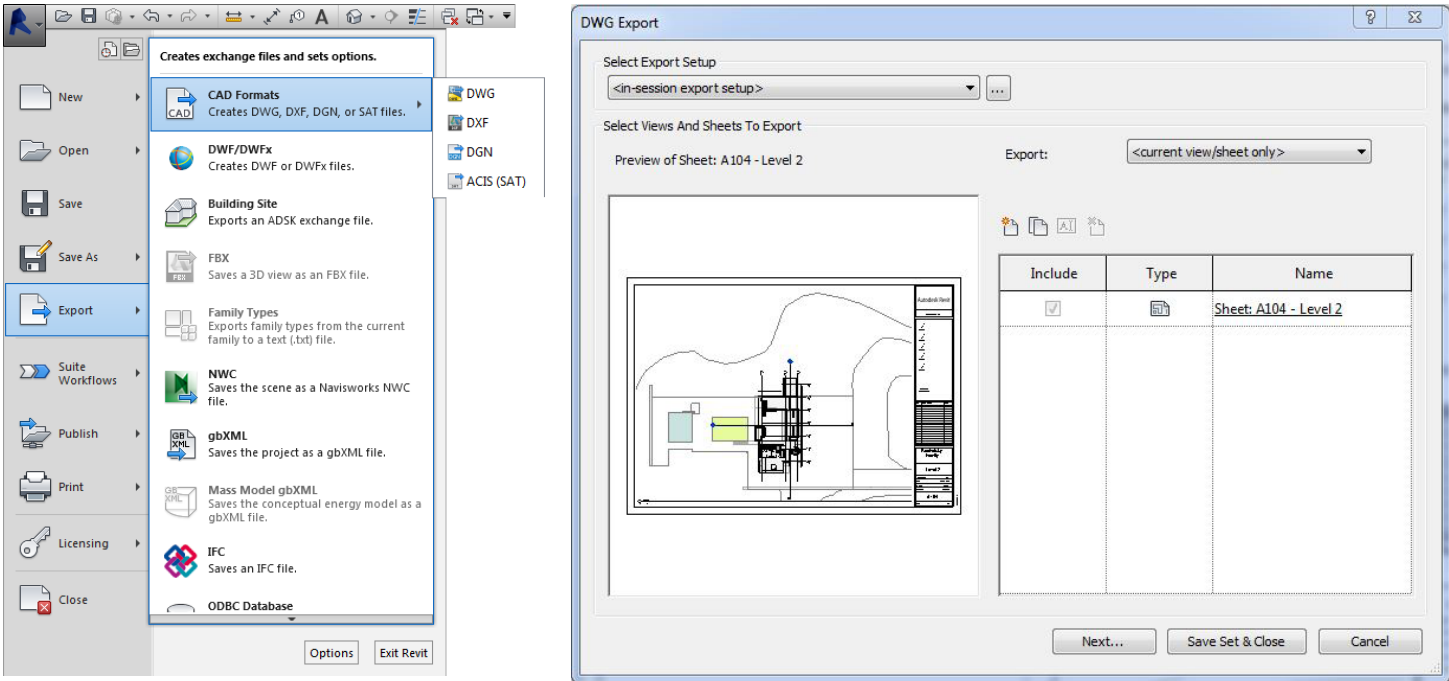
Adding sheets

The most organized and fastest way to print is to prepare a **set of sheets**. In the Project Browser there is a group that says Sheets. You can create new sheets easily. Revit will ask you to select a title block. If you have filled in the Project Information, the title block will be filled in automatically with this information. If you haven't filled in that information, you can do it manually. When you place a view on a sheet Revit automatically fills in the **references**. It numbers the drawing, and it fills in the references to the sheet. To add a view to the sheet there's a couple ways to do it. The first way is to simply drag and drop it from Project browser and place that view where you want, or you can use a **dedicated tool** in the Ribbon to do it.



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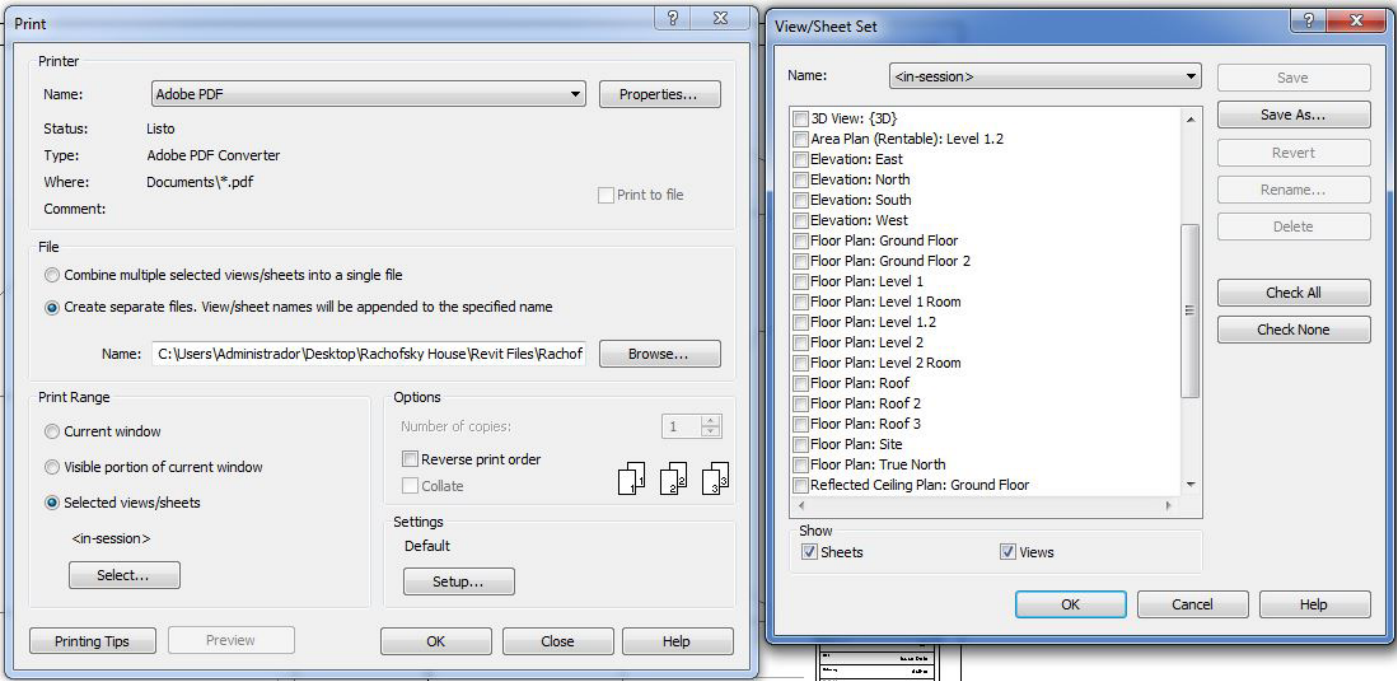


Printing can be also very fast in Revit. You can select various Sheets or Views and send them all to plot at once. For the Print range area you have three options: Current window, Visible portion of the current window and Selected views/sheets. If you choose Selected views/sheets, it will give you the complete list of views and sheets in the project and you can choose which ones you want to print and give that set a name. This is how you can print them all at once. You will be able to access that selection later by choosing that set.

Exporting to AutoCAD

As popular as Revit is, AutoCAD is still very prevalent in the building industry. As a result, there may be times when you will have to export your project to AutoCAD. Revit is capable of exporting directly to AutoCAD's native DWG format, to DXF, Microstation DGN, and even Asics 3D SAT files. The process is always similar. To export your Revit project to DWG, you can choose from the available viewsor sheets in your project. In the export setup dialog we can map the objects in Revit to layers in AutoCAD, so the AutoCAD drawing will have exactly the layers that we want and we won't have to do any manual work.

Printing and creating a PDF



We hope to have given you an idea of that BIM/Revit can offer you. Hopefully we were able to communicate our passion for this process and this tool and you can start experiencing the benefits our using it in your everyday life. Remember, these are some benefits of working in BIM:

reacts better to change.

B

saves time and money.
Improves Productivity.

offers new services.
Reduces Conflicts
& Changes

M

Reduces errors.

BIM IS THE PRESENT.
BIM IS THE FUTURE.

WHY S-BIM IN THE SCHOOL OF ARCHITECTURE?

S-BIM teaching in Architecture and Building Engineering degree permits the inclusion of building structure without losing the interferences with architecture and MEP. It is possible to use a single 3D model valid in other workshops with analysis and parallel developments. Furthermore, the throughput improvement allows the control of subjects until now managed by other specialties such as descriptions, time schedules, budgets or sustainability. BIM has a great facility for incorporating current methods of representation like parametric design, global project management systems, or the inclusion of international standards. The overall building management system will have a BIM methodology in the medium term. At workshops, students understand structural BIM as part of the design in the creation of the building and not as an auxiliary discipline for the construction project.

Oscar Liébana

Founder of BIM Lab UEM



3D+4D+5D

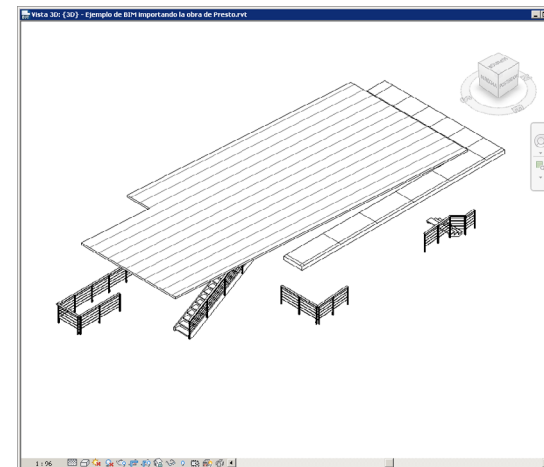
Execution plan drawings are just a first result of **BIM usage**. From the digital model is easy to quantify the objects within, obtaining a detailed list of every single element with its geometric dimensions, organized by category, types, levels and zones. If we relate every **type** with its correspondent field task, prices, descriptive texts and other associated information is obtained. **Quantification** is thereby transformed in the necessary budget (5D) for the project appraisal and execution contracting.

A BIM element is identified by a unique code, therefore the 3D model and the budget are **synchronized** and can reflect project changes in a bidirectional through the whole building life-cycle. The **model** is also useful as a baseline for planning and results and reports visualization. Several commercial systems allow a budget reclassification for a basic structure plan or even for a 3D model direct extraction. While including times and precedences, project management software can calculate time and date estimations. This results can be imported into the BIM model for a **dynamic execution process visualization**, always under the systems possibilities.

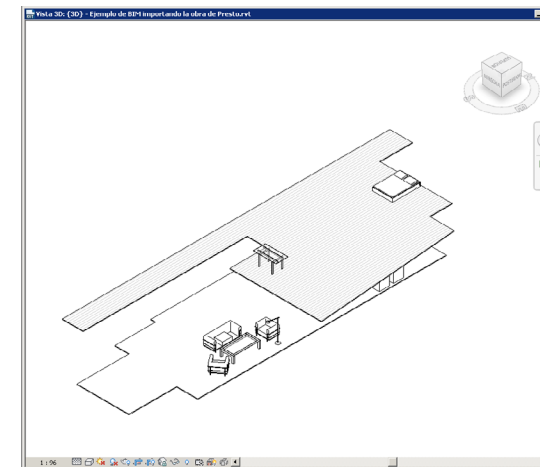
Fernando G. Valderrama

Presto Authors

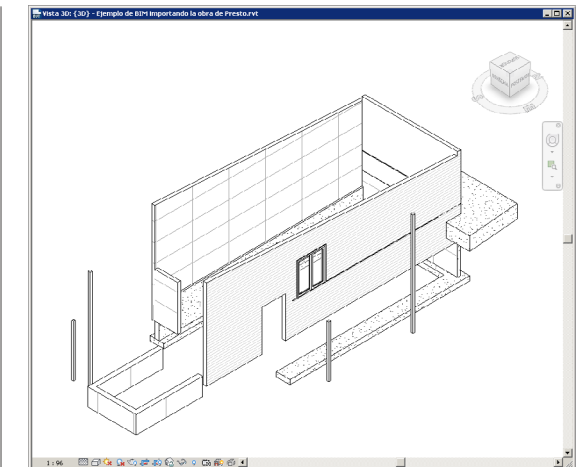
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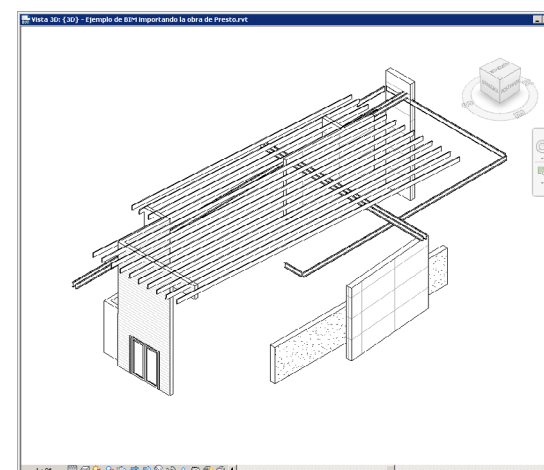
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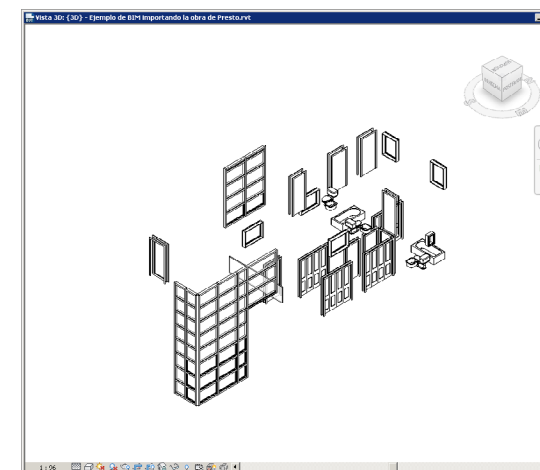
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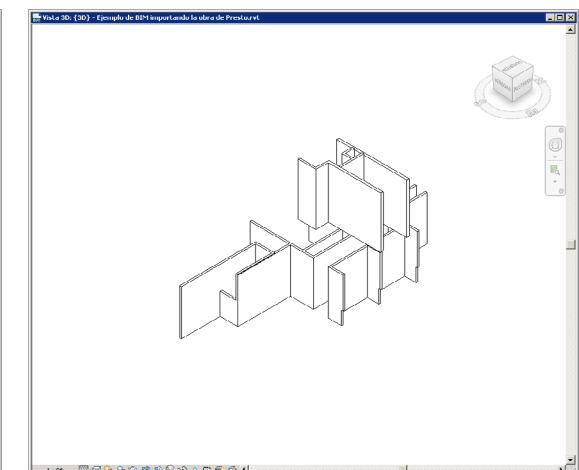
Phase 4



Phase 5



Phase 6



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BIM BENEFITS FOR BUILDING OWNER ON FACILITY MANAGEMENT

After the revolutionary emergence of BIM during the past decade as a new technology for the building industry, its benefits have been arousing owners interest. Accurate, timely and relevant information is centralized giving everyone access to the same data and greatly improving communication and collaboration for the project team.

BIM best values for facility management

Improved space management: Facility professionals can reduce vacancy and ultimately achieve major reductions in real estate expenses by understanding the details of how space is used. Efficiently used space management can be quickly visually shown. **Streamlined maintenance** Entering the product and asset information required for preventive maintenance is the key challenge in developing a maintenance program. The Information about building equipment stored in BIM models can eliminate months of effort to accurately populate

Efficient use of energy. BIM can help facilitate the analysis and comparisons of various energy alternatives to help facility managers dramatically reduce environmental impacts and operating costs. Regarding to LEED-EBOM certification and recertification, the BIM model is the first stop shop to identify which credits to tackle or which ones should be improved upon. **Economical retrofits and renovations.** A “living” BIM model provides an easier means of representing three-dimensional aspects of the building. Better information about existing conditions reduces the cost and complexity of building renovation and retrofit projects.

Enhanced lifecycle management. Some building design professionals are embedding data on life expectancy and replacement costs in BIM models, thereby helping an owner understand benefits of investing in materials and systems that may cost more initially but have a better payback over the life of the building. Integrating a **BIM model** with a facility’s maintenance management system can automate preventive maintenance program while connecting with the software package ensuring an even more robust maintenance program. BIM model becomes immediately in an electronic owners manual. After all, nothing has more value than information, as the common saying goes.

There are several ways of obtaining a BIM model of the building for facility management. The **best practices** are: populating the model while project and construction development where architects and contractors work through the **IPD** (Integrated Project Delivery) workflow in order to obtaining owner’s final BIM model delivery. A BIM consulting firm may build up the BIM model for facility management purpose. BIM is **not restricted** for new project buildings, but can be developed for existing ones.

David I. García

CEO in BaboonLab
FM:Systems Iberia busnisses partner



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WHY SHOULD YOU AND YOUR OFFICE MAKE THE MOVE TO BIM AS SOON AS POSSIBLE?

**Think no more.
The future is now.**

BIM is not some passing fad. BIM is not some new trending topic that will vanish with time. BIM is here and it is here to stay.

BIM is not some passing fad. BIM is not some new trending topic that will vanish with time. BIM is here and it is here to stay. If you are a professional related to the AEC industry you should start realizing that the processes of designing and building are progressing by **leaps** and **bounds**. We must all adapt to the new methods, procedures and techniques. In the particular case of an architectural practice, most professionals work from a craftsmanship point of view which is not appropriate, and have a very archaic documentation production scheme. One thing should be clear. BIM processes **do not** affect our ways of designing. In any case, they help our design workflows.

What BIM processes really affect and transform, as a new and powerful tool, are all our procedures of documentation, production and collaboration methods. Here is where a firm commitment to a **strong implementation** of these new processes and technologies can make your office stand out from the rest. Not only the fact that you use BIM technology will be an asset to your office. With these new tools you will, among other things, have **greater control** over your project, avoiding lack of consistency among documents.

You will have a tool that allows you to manage changes very **efficiently** and you will have a much greater capacity to easily generate valuable documentation than by using traditional design methods based in CAD. All this is possible with a great effort. It takes time and hours of study by all parts involved, but once you have delivered your first project using BIM technology I can assure you that you **will never** want to go back.

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Revit Consultant and Specialist
Revit Architecture Certified Professional**



Autodesk Wikihelp

Mastering Autodesk Revit Architecture 2014. James Vandezande, Eddy Krygiel, Phil Read

Paul F. Aubin <http://paulaubin.com/> <http://www.lynda.com>

Brian Myers <http://www.lynda.com>

Shades of Grey. Andy Milburn <http://grevity.blogspot.com.es/>

What Revit Wants <http://whatrevitwants.blogspot.com.es/>

Revit Op Ed <http://revitoped.blogspot.com.es/>



COLORPHONE

Title
Typeface
Heads
Body
Date

Revit
Dinpro
Pt Sans Narrow
2013/2014

